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Research Paper

Transient analysis of megawatt-level space gas-cooled reactor coupled with He-Xe Brayton cycle system

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ABSTRACT

The development of the aerospace industry and advancements in aerospace technology have created an urgent need for space power supply. A self-developed thermal hydraulic system analysis program is utilized to examine the transient response safety of gas-cooled reactors and investigate the performance of the coupled Brayton cycle overall system. The typical transient accident conditions were simulated, including positive reactivity insertion, turbine-alternator-compressor shaft speed drop, and loop coolant flow rate reduction. By considering cladding integrity as the safety criterion, it is determined that the reactor studied in this work has a safety margin of 0.152 $ when introducing reactivity. The introduction of positive reactivity enhances reactor power output and heat generation. As more heat is carried away by He-Xe gas into the turbine, both operating temperature and heat transfer efficiency within the thermal cycle system are further improved. With an introduction of 0.13 % positive reactivity into the core, reactor thermal power can be stabilized at 110 % of its rated value while increasing system thermoelectric conversion efficiency by 2.7 %. During a continuous turbine-alternator-compressor shaft speed drop incident (from 100 % to 80 % rated speed), the total electrical power drops to 580 kWe along with reductions in temperature ratio and pressure ratio for both turbine and compressor. System cycle efficiency shows a positive correlation with turbine-alternator-compressor shaft speed within an appropriate range. In a sudden coolant flow rate drop incident, when the loop flow rate decreases to 60 % of its rated value, both reactor power and turbine-alternator-compressor shaft power decline significantly: the total electrical power drops from 1096 kWe to 216 kWe while the system's cycle efficiency decreases from 32.1 % to 12.13 %. This work provides valuable insights into conceptual design considerations as well as safety assessment aspects pertaining to megawatt-class space.

1. Introduction

An efficient energy system is fundamental for deep space exploration missions. Nuclear energy, characterized by high energy density and long lifespan, offers an effective solution for future deep space energy challenges. For megawatt-level deep space missions, Space Nuclear Reactor Power Systems (SNRPS) will be the only viable option [1]. SNRPS primarily comprises three components: a nuclear reactor heat source, a thermoelectric conversion system, and a thermal radiation system for waste heat rejection. Based on reactor cooling methods, these can be classified into alkali metal heat pipe cooling, liquid metal cooling, and inert gas cooling [2]. Energy conversion technologies are divided into static and dynamic types. Static energy conversion includes thermoelectric generation, magnetohydrodynamic generation (MHD), thermionic conversion, alkali metal thermal-electric conversion (AMTEC), and thermophotovoltaic conversion. Dynamic energy conversion encompasses the Brayton cycle [3], Rankine cycle [4], and Stirling cycle [5].

The current preferred choice for power systems in space reactors is dynamic energy conversion methods. In 2002, NASA introduced the "Prometheus Project" [6], which designed a 200 kWe power generation system based on a high-temperature gas-cooled reactor that utilizes a He-Xe closed Brayton cycle for thermoelectric conversion. Within the framework of the Prometheus Project, NASA developed the Jupiter Icy Moons Orbiter (JIMO) mission [7].

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In 2009, the Russian Federal Atomic Energy Agency proposed the development of a megawatt-class deep space nuclear-powered spacecraft. The TEM project utilizes the Brayton cycle as the thermoelectric conversion system with He-Xe as the working fluid. The outlet temperature of the reactor coolant is approximately 1500 K, and it has a power generation capacity of 1000 kWe. The system incorporates droplet radiative cooling to achieve an ideal heat exchange efficiency of 34 % [8]. Russia's 3.4 MWt nuclear thermal-electric dual-mode rocket engine employs H_2 as propellant for propulsion mode, while in power generation mode, it utilizes the He-Xe Brayton cycle along with a backup dual Brayton thermoelectric conversion system [9].

In the field of space reactor technology, research directions primarily encompass the design of space reactor systems, simulation and validation of reactors, as well as the coupling and simulation of space reactors with power generation systems. El-Genk et al. [10] developed the Thermionic Fuel Element simulation model to conduct comprehensive studies on reactor startup and transient operating conditions. NASA's Glenn Research Center (GRC) devised the Closed-Loop System Simulation model for simulating transient system operations, which was validated through experiments utilizing the Brayton Power Conversion Unit (BPCU) [11].

In the 1980s, NASA and the Navy collaborated on the Combined Cycle Engine Program (CCEP), wherein they developed a simulation code for assessing the off-design performance of a Brayton cycle power generation system based on a space gas-cooled reactor [12]. Sandia National Laboratories developed a MATLAB/Simulink-based transient simulation code to model the dynamic behavior of gas-cooled space reactors and metal-cooled reactor systems with Brayton cycles [13]. The focus of this study is to develop and validate an integrated dynamic system model, thus the generated closed-loop Brayton cycle power is relatively small, ranging from 10 to 30 kWe, not reaching the megawatt level. Wu et al. [14] established a dynamic model of the He-Xe Brayton cycle LCSR using the RESYS reactor system code to simulate steady-state operation as well as four typical unexpected scenarios. It should be noted that the simulated reactor object he established is a lithium-cooled reactor rather than an gas-cooled one. Wang et al. [15] developed a dynamic simulation program for space gas-cooled reactors combined with CBC. The study primarily confirms that even with a reactivity of 0.4 $ introduced into the reactor core during full power operation in steady state, there still exists a significant safety margin for hot spots, thus this reactor has the potential to achieve higher steady-state design power. However simulations have not yet been conducted to assess its response under other transient typical accident conditions in this work. Zhang et al. [16] developed SNRPS analysis software for megawatt-class gas-cooled reactors combined with CBC. The primary objective of this study was to simulate and analyze mechanical failures associated with MFOBL1 and RIA accidents, while also verifying the system's self-regulating capability during accident scenarios. However, a comprehensive investigation into the response characteristics of the system was not conducted, and a baseload space gas-cooled reactor was utilized. Qin et al. [17] developed the SAMER system analysis program for open megawatt gas-cooled space nuclear reactors to analyze transient thermal hydraulic responses URIA and unprotected partial coolant LOFA. The focus of its research is mainly on the response of reactor systems, and there has not been a thorough analysis of the response laws for coupled Brayton cycles and overall performance efficiency.

Fig. 1 shows the previous research progress in this work. The availability of simulation programs for space gas-cooled reactor coupled He-Xe Brayton cycle system is limited, with most of them operating at power levels below a megawatt. Existing transient analysis work on this system primarily focuses on the reactor module, exploring its safety boundaries without protective measures, while neglecting to provide detailed insights into the response of the complete coupled Brayton cycle. Furthermore, there is a lack of in-depth analysis regarding changes and causes affecting the overall performance efficiency of the system when transitioning from transient conditions back to steady state. This study aims to address these research gaps by conducting a custom simulation program-based transient safety analysis on megawatt-class space nuclear reactors. In addition to exploring the safety boundaries of the reactor, this study also simulates accident scenarios under protective measures. The research findings will facilitate operational analysis and evaluation of control strategies for nuclear reactors equipped with Brayton cycle conversion systems, as well as aid in modeling other megawatt-class space gas-cooled reactor systems.

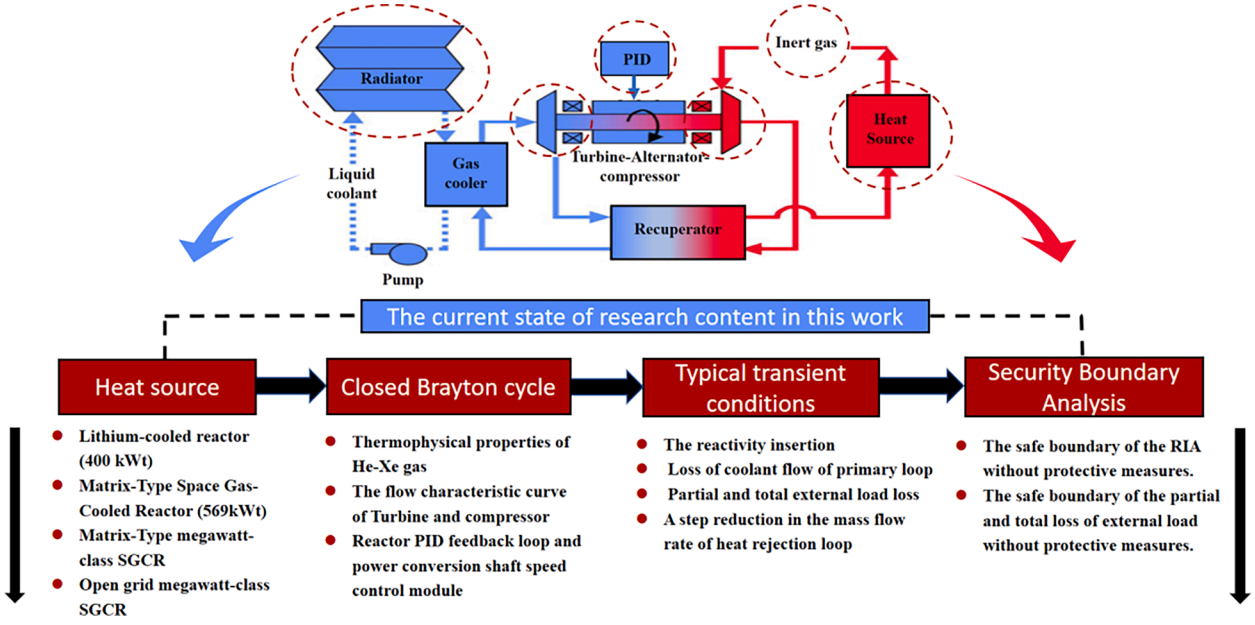


Fig. 1. The previous state of research content in this work.

Nomenclature

Symbol	
A	Flow area of the coolant channel
D_e	Equivalent diameter of the coolant channel
α_i	Temperature coefficient of material i
f	Frictional resistance coefficient
h_g	Enthalpy of He-Xe gas
m	Mass flow
N_{shaft}	Shaft speed of the TAC
T	Temperature
T_U	Temperature of the fuel core
T_C	Cladding temperature
T_{iavg}	Average temperature of material i
T_{iref}	Reference temperature of material i
P	Pressure
P_{shaft}	Shaft power of the TAC
P_{turb}	Turbine power
P	Pressure
P_g	Pressure of He-Xe gas
P_F	Core fission power
P_D	Total decay power of the reactor
ρ	Total reactivity
ρ_0	Initial reactivity
ρ_g	Density of He-Xe gas
P_{ex}	External reactivity introduced
β_i	Fraction of the ith group of delayed neutrons
δ_L	Thickness of the inner liner
δ_C	Cladding thickness
η_{sC}	Isentropic efficiency of the compression process
η_{pC}	Isothermal efficiency of the compression process
η_{sT}	Isentropic efficiency within the turbine
η_{pT}	Isothermal efficiency within the turbine

2. Brief description of the high temperature gas cooled space reactor

2.1. He-Xe gas-cooled reactor

The gas-cooled reactor coupled Brayton cycle system is illustrated in Fig. 2, while the reactor core structure is depicted in Fig. 3. The inlet gas flows along the pressure vessel wall, which consists of Nickel-Chromium-Iron alloy 718. The reactor core comprises an upper grid plate, a lower grid plate, 732 fuel elements, and 13 safety control rods. The fuel elements are securely fastened to the grids on the upper and lower grid plates with compression grids assisting in their fixation. The fuel elements are arranged in a triangular pattern within the core with a spacing of 14.2 mm. The control rods contain B_4C pellets enclosed within a 1 mm thick outer shell. A metallic rubber layer between the shell and B_4C pellets compensates for radial expansion of B_4C. The radiation shielding cavity of the reactor is filled with components made of LiH.

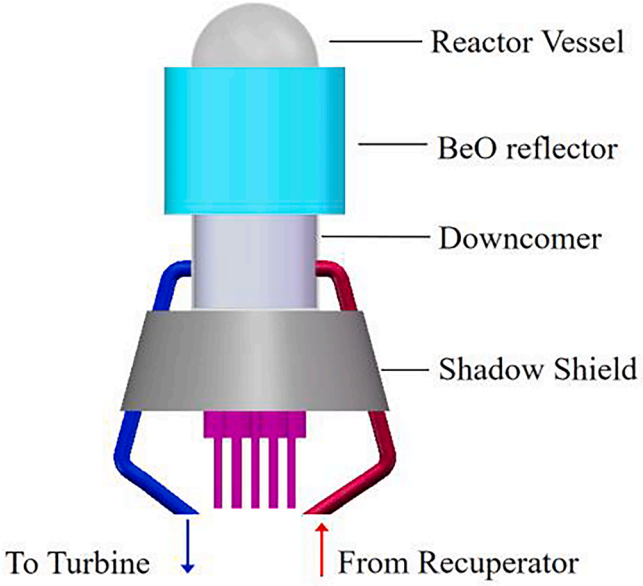


Fig. 3. Space gas-cooled reactor structure.

The fuel structure of the open grid gas-cooled reactor is illustrated in Fig. 4. The fuel element comprises a fuel region and a supplementary space. Within the fuel region, there are several components including a cladding for the fuel element, axial reflector layers at both cold and hot ends, UO_2 fuel pellets, flat baffles, and exhaust channels in the form of ducts measuring 2.9 × 0.2 mm. The UO_2 fuel pellets are disk-shaped with sintered material and feature an internal central hole. The cladding is composed of Mo-Nb-Zr alloy. The design parameters of the space gas-cooled reactor as shown in the Table 1.

2.2. Recuperated closed Brayton cycle

Considering the stringent volume and mass requirements in space missions, it is preferable to utilize a binary He-Xe mixture with a molar mass ranging from 40-45 $g{\cdot}mol^{-1}$ [18] instedad of pure Helium. The mole fraction of helium varies between 60–90 %, thereby minimizing system component mass and size while meeting compactness criteria.

Fig. 5 depicts the schematic diagram of a closed Brayton cycle system. This closed-loop cycle involves gas circulation through the

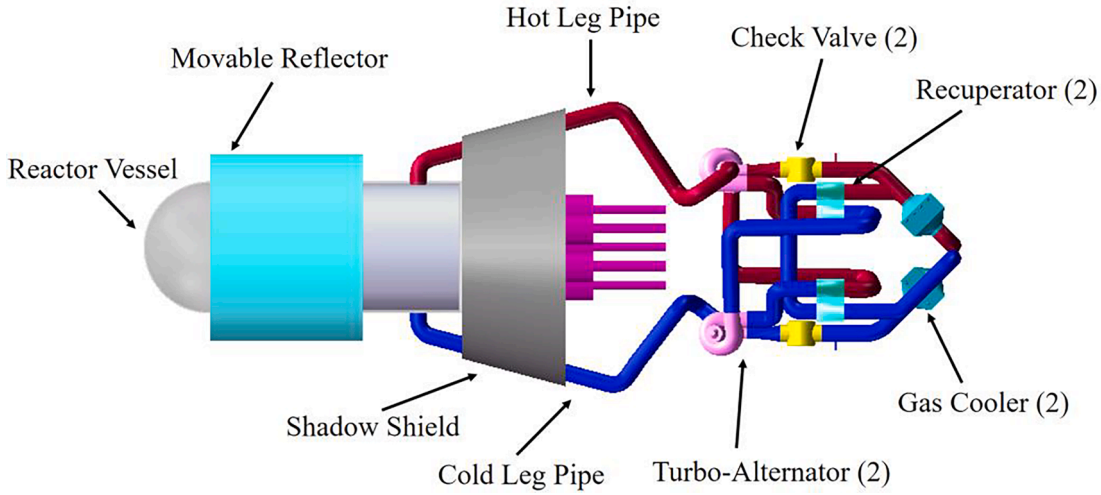


Fig. 2. The cross-section of the gas-cooled reactor coupled Brayton cycle system.

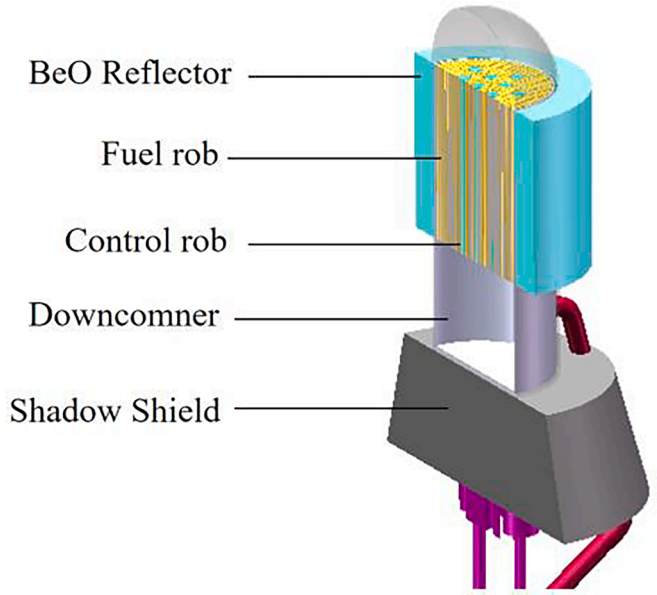


Fig. 4. Fuel structure of open grid space gas- cooled reactor.

Table 1
The design parameters of the space gas-cooled reactor.

Parameters	Value	Parameters	Value
Neutron spectrum	Fast neutron reactor	Radial reflector thickness(mm)	103
Thermal power(MW)	3.4	Fuel rod amount	732
Coolant mass flow rate(kg/s)	14.236	Control rod amount	13
Control rod guide tube outer diameter(mm)	35	Pitch(mm)	14.2
Core diameter(mm)	416	UO_2 fuel pellet inner diameter(mm)	3.0
Downcomer inner diameter (mm)	444	UO_2 fuel pellet outer diameter(mm)	10.9
Downcomer outer diameter (mm)	464	He-Xe gap thickness (mm)	0.05
Pressure vessel thickness (mm)	6	Cladding thickness (mm)	1.0
Pressure vessel outer diametr(mm)	476	Active zone length	450

compressor and recuperator, entry into the reactor, passage through the turbine, re-entry into the recuperator, and ultimately reaching the waste heat rejection system (space radiator). The reactor is positioned on the left side of the diagram while the waste heat rejection system is located on its right side.

The T-S diagram in Fig. 6 depicts the temperature-entropy characteristics of a single-loop He-Xe Brayton cycle. The pressure ratio across the primary compressor (from inlet state 1 to outlet state 2) is approximately 2.6:1, resulting in an associated increase in pressure. As the gas undergoes compression, its temperature also elevates, leading to a corresponding rise in temperature at state 2.

$$\frac{T_{o2}}{T_{o1}} = 1 + \frac{\left(p_{o2}/p_{o1}\right)^{\frac{\gamma-1}{\gamma}} - 1}{\eta_{sC}} = \left(\frac{p_{o2}}{p_{o1}}\right)^{\frac{\gamma-1}{\gamma}/\eta_{pC}} \tag{1}$$

The subscript "o" denotes total temperature or total pressure, Where Subscripts "1-6″ correspond to the process quantities in Fig. 6, respectively. $\gamma = c_p / c_v$ denotes the ratio of specific heats at constant pressure to constant volume, specifically for an ideal gas. The definitions for total temperature and total pressure are as follows:

$$T_o = 1 + \frac{v^2}{2C_p T_s} \tag{2}$$

$$p_o = p_s\left(\frac{T_o}{T_s}^{\frac{\gamma-1}{\gamma}}\right) \tag{3}$$

The subscript "s" represents the static values of temperature and pressure. The work performed during the compression process is given by: $W_{comp} = m{\cdot}c_p(T_{o2} - T_{o1})$. Processes 2 to 3 and 5 to 6 represent the entropy changes across the recuperator. During process 2 to 3, the compressed He-Xe gas is heated from T_{o2} to T_{o3} by the high-temperature exhaust gas exiting the turbine at the recuperator. The high-temperature gas leaving the turbine is cooled from T_{o5} to temperature T_{o6}. The heat transfer in each regenerator channel can be expressed by: $Q_{23} = m{\cdot}c_p(T_{o3} - T_{o2})$.

Under steady-state conditions, the thermal energy gained in the low-temperature section must precisely match the thermal energy lost in the high-temperature section. Therefore, $Q_{23} = Q_{56}$. Thus $(T_{o3} - T_{o2}) = (T_{o5} - T_{o6})$, the efficiency of the regenerator can be equivalently expressed as:

$$\varepsilon_x = \frac{T_{o5} - T_{o6}}{T_{o3} - T_{o2}} \tag{4}$$

The expansion process within the turbine is represented by processes 4 to 5. The pressure drop during this expansion process is approximately equal to the pressure increase within the compressor. The gas expansion within the turbine generates mechanical work, given by: $W_{45} = m{\cdot}c_p(T_{o4} - T_{o5})$. The temperature at the turbine outlet depends on the inlet temperature and pressure.

$$\frac{T_{05}}{T_{04}} = 1 - \eta_{sT}\left(1 - \left(p_{05}/p_{o4}\right)^{\frac{\gamma-1}{\gamma}}\right) = \left(\frac{p_{05}}{p_{o4}}\right)^{\frac{\gamma-1}{\gamma}\cdot\eta_{pT}} \tag{5}$$

Subsequently, the gas in the turbine undergoes expansion work and then enters the high-temperature region of the recuperator, where it engages in thermal exchange with the He-Xe gas from the compressor outlet on the low-temperature side. The outlet temperature is determined by Eq.(5).

The final segment of the cycle (processes 6 to 1) involves the waste heat dissipation system (gas cooler). Excess heat generated by the Brayton cycle is transferred from the He-Xe gas to water in the cooler, conveying the thermal energy from the Brayton cycle to the cooling system. Ultimately, this heat is emitted into space through thermal radiation via the radiator [19].

3. Model

3.1. Reactor kinetics model

3.1.1. Point kinetics reactor model

The reactor power is composed of two main components: fission power and decay power. This study presents an analysis of variations in core fission power using point reactor kinetics equations that incorporate six groups of delayed neutrons, while also accounting for reactivity feedback from various mechanisms. The formulation of the point reactor kinetics equations can be described as follows:

$$\begin{aligned}\frac{dP_F(t)}{dt} &= \frac{\rho(t) - \sum_{i=1}^{6}\beta_i}{\Lambda}P_F(t) + \sum_{i=1}^{6}\lambda_i C_i(t)\\ \frac{dC_i(t)}{dt} &= \frac{\beta_i}{\Lambda}P_F(t) - \lambda_i C_i(t) \quad i = 1, 2\cdots, 6\end{aligned} \tag{6}$$

Where: P_F — Core fission power (W); t — Time (s); ρ — Total reactivity ($); β_i — Fraction of the ith group of delayed neutrons; Λ— Neutron

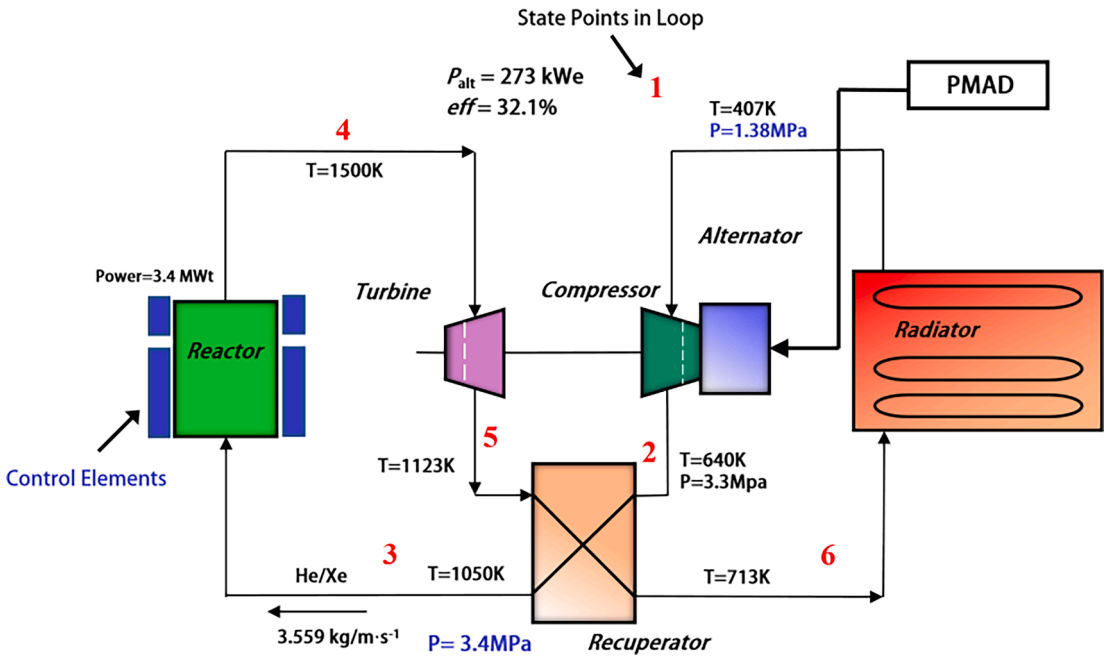


Fig. 5. Schematic diagram of gas-cooled reactor coupled reheated Brayton cycle.

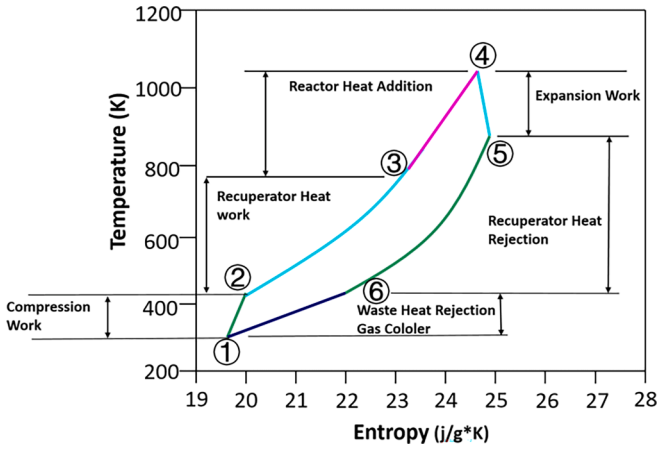


Fig. 6. Temperature entropy diagram of a single-loop He-Xe Brayton cycle.

generation time (s); λ_i— Decay constant of the i group of delayed neutrons (s^{-1}); C_i — Concentration of the i group of delayed neutron precursors (m^{-3}).

When calculating decay power, it is assumed that all fuel elements within the reactor experience identical variations in decay power and that multiple groups of fission products undergo decay processes. Following simplification, the formula for calculating decay heat can be obtained as follows:

$$\frac{dP_{Dj}(t)}{dt} = E_j P_F(t) - \lambda_j P_{Dj}(t) \quad j = 1,2,3,4$$
$$P_D(t) = \sum_{j=1}^{4} \lambda_j P_{Dj}(t) \tag{7}$$

The total reactor power can be expressed as:

$$P_{Eff}(t) = \left(1 - \sum_{j=1}^{4} E_j\right) P_F(t) + P_D(t) \tag{8}$$

Where: $P_{D,j}$ — Decay power of the j group of fission products (W); E_j — Effective energy fraction of the jth group of fission products; λ_j— Decay constant of the j group of delayed neutrons (s^{-1}); P_D — Total decay power of the reactor (W); P_{Eff} — Effective power of the reactor (W).

3.1.2. Feedback mechanisms and models

In a gas-cooled reactor, apart from accounting for the external reactivity introduced by mechanical means such as control drum rotation, lateral reflector movement, and control rod insertion and withdrawal, due consideration should also be given to the effects of reactivity feedback. The total reactivity of the core can be mathematically expressed as:

$$\rho(t) = \rho_0 + \rho_{ex}(t) + \sum_i \rho_i(t) \tag{9}$$

Where: ρ_0 — Initial reactivity ($), which must be 0 in steady-state; ρ_{ex} — External reactivity introduced by control mechanisms such as control drums or control rods ($); ρ_i — Various reactivity feedbacks within the reactor ($).

In a space gas-cooled reactor, the considered reactivity feedback effects encompass the Doppler effect of the fuel, fuel cladding, core matrix (in matrix-type gas-cooled reactors), temperature reactivity feedback of the pressure vessel and lateral reflector, as well as void coefficient reactivity feedback of the gaseous coolant. The temperature reactivity feedback of each material can be mathematically expressed as:

$$\rho_i(t) = \alpha_i\left(T_{iavg}(t) - T_{iref}\right) \tag{10}$$

Where: α_i — Temperature coefficient of material i ($\cdot$K^{-1}); T_{iavg} — Average temperature of material i (K); T_{iref} — Reference temperature of material i (K).

3.2. Heat transfer model

3.2.1. He-Xe gas flow and heat transfer

In the system of the space gas-cooled reactor, the working fluid He-Xe gas remains in the gaseous phase, exhibiting single-phase flow. In a gravity-free environment such as space, kinetic energy changes and axial heat conduction of the fluid are disregarded. Taking into account the compressibility of the He-Xe gas, the following control equations can be derived.

Mass conservation equation:

$$\frac{\partial \rho_g}{\partial t}+\frac{\partial}{\partial z}\left(\frac{W_g}{A}\right)=0 \tag{11}$$

Momentum conservation equation:

$$\frac{\partial}{\partial t}\left(\frac{W_g}{A}\right)+\frac{\partial}{\partial z}\left(\frac{W_g^2}{\rho_g A^2}\right)=-\frac{\partial p_g}{\partial z}-\frac{fW_g|W_g|}{2D_e\rho_g A^2} \tag{12}$$

Energy conservation equation:

$$\rho_g\frac{\partial h_g}{\partial t}+\frac{W_g}{A}\frac{\partial h_g}{\partial z}=\frac{q_w\Pi}{A}+\frac{\partial p_g}{\partial t} \tag{13}$$

Where: ρ_g — Density of He-Xe gas (kg·m^{-3}); z — Coordinate along the flow direction (m); W_g — Mass flow rate of He-Xe gas (kg·s^{-1}); h_g — Enthalpy of He-Xe gas (J·kg^{-1}); q_w — Heat flux density at the coolant channel wall (W·m^{-2});Π— Heating perimeter of the coolant channel (m).

When solving the temperature distribution using Newton's law of cooling, it is essential to have knowledge of the convective heat transfer coefficient, which is typically determined by Nu. Nu is commonly calculated through an empirical relationship. However, due to the small Pr of He-Xe gas, its empirical relationship differs from that of general gases. Based on experimental research in literature [20,21], for calculating core convective heat transfer, both wall fluid viscosity and main fluid viscosity are considered according to the Gnielinski formula [22], with reference made to the formula utilized by RELAP5-3D in order to provide a more precise determination of Nu.

$$\begin{aligned} Nu&=\frac{(\xi/8)RePr}{K_1(\xi)+K_2(Pr)\sqrt{(\xi/8)}(Pr^{2/3}-1)}\cdot\left(\frac{\mu_f}{\mu_w}\right)\\ \xi&=(1.82\log Re-1.64)^{-2}\\ K_1(\xi)&=1+3.4\xi\quad K_2(Pr)=11.7+1.8Pr^{-1/3}\end{aligned} \tag{14}$$

The formula utilized in the Recuperator primarily references the Promethus report and the Kays formula [23]:

$$Nu=0.09Re^{0.58}\mathrm{Pr}^{1/3} \tag{15}$$

The Eq. (14) is used for calculating Nu in the Gas Cooler when Re > 10^4, the Eq.(15) employed for Re < 10^4.

3.2.2. Core conduction

The fuel element structure of the Gas-Cooled Reactor (GCR) is illustrated in Fig. 7. It comprises a cylindrical UO_2 fuel core, followed by a fission gas gap and an Nb-Zr cladding arranged from innermost to outermost. Surrounding the cladding is an equivalent annular coolant.

(1) Control equation for the nodes within the fuel core:

$$\rho_U c_{p,U}\frac{\partial T_U}{\partial t}=\frac{1}{r}\frac{\partial}{\partial r}\left(\lambda_U r\frac{\partial T_U}{\partial r}\right)+Q_U \tag{16}$$

Where: T_U — Temperature of the fuel core (K); r — Radial position (m);λ_U — Thermal conductivity of the fuel core (W·m^{-1}·K^{-1}); Q_U — Volume heat source density of the fuel core (W·m^{-3}).

(2) Thermal balance equation for the Re liner:

$$\begin{aligned}\rho_L c_{p,L}\frac{\partial T_L}{\partial t}&=\frac{\Pi_L}{A_L}\frac{\lambda_G}{\delta_G}(T_U-T_L)+\frac{\Pi_U}{A_L}\varepsilon_{UL}\sigma(T_U^4-T_L^4)\\&-\frac{\Pi_{CI}}{A_L}\left[(T_L-T_C)/\left(\frac{\delta_L}{2\lambda_L}+\frac{\delta_C}{2\lambda_C}\right)\right]\end{aligned} \tag{17}$$

Where: A_L — Cross-sectional area of the inner liner (m^2);Π_{CI} — Perimeter of the inner surface of the cladding (m); T_C — Cladding temperature (K);δ_L — Thickness of the inner liner (m);δ_C — Cladding thickness (m);λ_C — Thermal conductivity of the cladding (W·m^{-1}·K^{-1}).

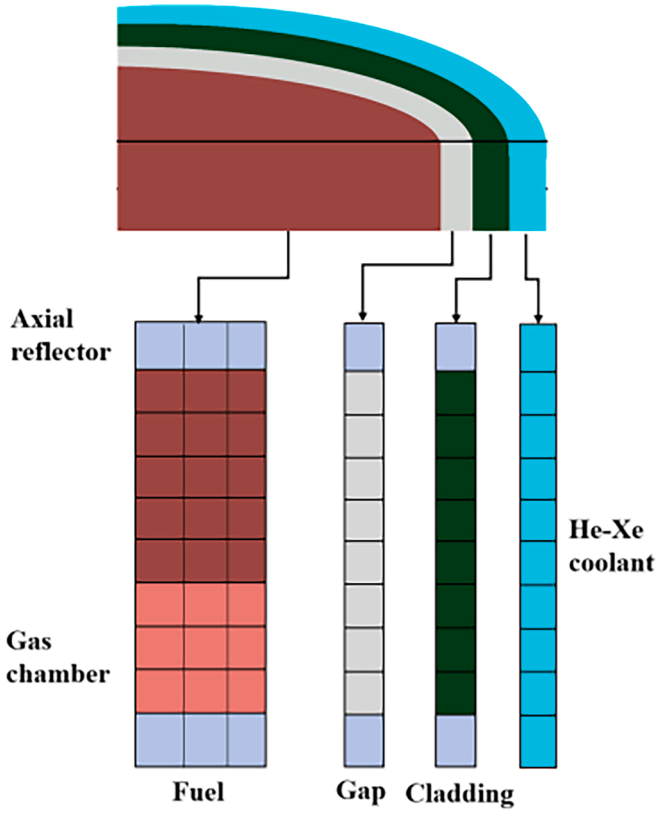


Fig. 7. Division of Control Body of Single-channel Fuel Element in Open Grid Gas-cooled Reactor.

The first term on the right-hand side delineates the heat conduction between the fuel core and the inner liner through the interposed gap layer. The second term describes the radiative heat transfer occurring between the core and the inner surface of the liner, while the third term denotes the heat conduction between the fuel liner and the cladding.

(3) Thermal balance equation for the fuel cladding:

$$\begin{aligned}\rho_C c_{p,C}\frac{\partial T_C}{\partial t}&=\frac{\Pi_{CI}}{A_C}\left[(T_L-T_C)/\left(\frac{\delta_L}{2\lambda_L}+\frac{\delta_C}{2\lambda_C}\right)\right]\\&-\frac{\Pi_{CO}}{A_C}h_{Cg}(T_C-T_g)\end{aligned} \tag{18}$$

Where: A_C — Cross-sectional area of the cladding (m^2);Π_{CO}— Outer surface perimeter of the cladding (m);ε_C— Cladding emissivity; T_g — He-Xe gas temperature (K).

The first term on the right side denotes the heat conduction between the fuel inner liner and the cladding, while the second term represents the convective heat transfer between the fuel cladding and the gas coolant.

3.3. Brayton cycle model

3.3.1. Turbine-Alternator-Compressor (TAC) model

In the Brayton cycle power generation system of the space gas-cooled reactor utilizing He-Xe as the working fluid, the turbomachinery primarily consist of a compressor and a turbine. An alternator is positioned between the turbine and the compressor. These three components are interconnected on the same shaft, forming a Turbine-Alternator-Compressor (TAC) integrated unit, as depicted in Fig. 8.

The Turbine-Alternator-Compressor (TAC) provides shaft power to maintain torque balance, thereby driving the coolant through the

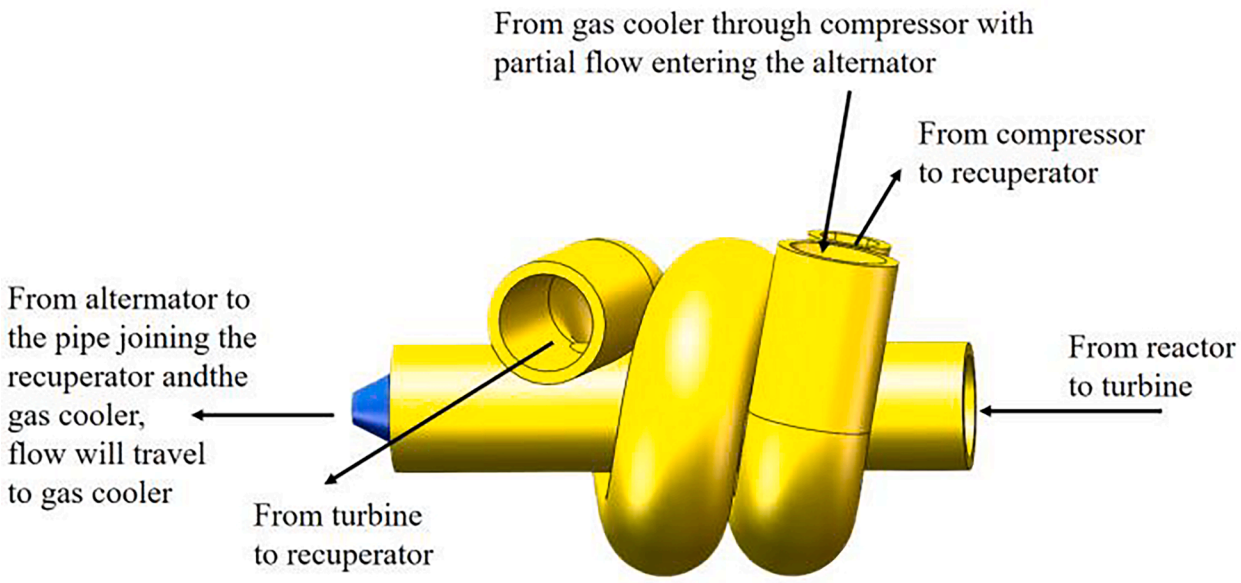


Fig. 8. Turbine-Alternator-Compressor (TAC) Model.

reactor. Consequently, the electrical load and operational parameters (temperature, pressure, flow rate, speed) of the turbine machinery, as well as the state of the gas Brayton cycle loop, directly influence the flow rate through the reactor, which in turn affects the reactor's power level. The core of the turbine machinery model resides in the shaft power balance of the TAC and the flow characteristic curves of the turbine compressor.

The Turbine-Alternator-Compressor (TAC) ensures torque balance by providing shaft power, thereby driving the coolant flow through the reactor. Consequently, the turbine machinery's electrical load and operational parameters (temperature, pressure, flow rate, speed), along with the state of the gas Brayton cycle loop, directly influence the reactor's flow rate and subsequently its power level. The core of the turbine machinery model lies in maintaining shaft power balance within the TAC and understanding the flow characteristic curves of the turbine compressor.

(1) Power Balance

The temporal derivative of shaft speed can be formulated as:

$$\frac{dN_{\text{shaft}}}{dt} = \frac{P_{\text{shaft}}}{I \cdot N_{\text{shaft}} \cdot 4\pi^2} \tag{19}$$

Power balance equation on the TAC shaft:

$$P_{shaft} = P_{turb} - P_{comp} - P_{alt} \tag{20}$$

Power consumed by the compressor:

$$\frac{p_{o4}}{p_{o5}} = f_{prT}\left(T_{o4}, p_{o4}, mdot, N_{rpm}\right) \Leftrightarrow mdot = f_{mprT}\left(T_{o4} \cdot P_{o4}, P_{o5}, N_{rpm}\right) \tag{21}$$

Power generated by turbine:

$$P_{turb} = m \cdot c_p \cdot (T_{o4} - T_{o5}) \tag{22}$$

Where: N_{shaft} — Shaft speed of the TAC (s^{-2}); P_{shaft} — Shaft power of the TAC (s^{-2}); I — Moment of inertia of the shaft ($kg \cdot m^2$); P_{turb} — Turbine power (W); P_{alt} —Alternator power (W); P_{com} — Compressor power (W).

(2) Flow Characteristic Curves

The flow characteristic curves employed in this study are derived from NREC's design specifications for a 132 kWe space gas-cooled reactor [13]. Dependent on temperature and pressure ratios, these curves establish correlations with factors such as inlet temperature, inlet pressure, mass flow rate, and shaft speed; thereby enabling accurate calculations for outlet temperature and pressure under specific inlet conditions. To precisely define these relationships within the dynamic model this work employ multivariable polynomial fitting techniques.

Compressor pressure ratio curve:

$$\frac{p_{o2}}{p_{o1}} = f_{prC}\left(T_{o1}, p_{o1}, m, N_{shaft}\right) \tag{23}$$

Compressor temperature ratio curve:

$$\frac{T_{o2}}{T_{o1}} = f_{TrC}\left(T_{o1}, p_{o1}, m, N_{shaft}\right) \tag{24}$$

Turbine pressure ratio curve:

$$\frac{p_{o4}}{p_{o5}} = f_{prT}\left(T_{o4}, p_{o4}, m, N_{shaft}\right) \Leftrightarrow p_{o5} = f_{prT}\left(T_{o4} \cdot P_{o4}, m, N_{shaft}\right) \tag{25}$$

Turbine temperature ratio curve:

$$\frac{T_{o4}}{T_{o5}} = f_{TrT}\left(T_{o4}, p_{o4}, m, N_{shaft}\right) \Leftrightarrow T_{o5} = f_{trT}\left(T_{o4} \cdot P_{o4}, m, N_{shaft}\right) \tag{26}$$

Where: f_{prC} — Compressor pressure ratio characteristic curve; f_{TrC} — Compressor temperature ratio characteristic curve; f_{prT} — Turbine pressure ratio characteristic curve; f_{TrT} — Turbine temperature ratio characteristic curve.

3.3.2. Heat exchanges module

Each Brayton cycle loop conversion system is equipped with a gas cooler (Fig. 9), which features a shell-and-tube counterflow heat exchanger. The shell side facilitates the flow of high-temperature He-Xe gas, while water serves as the cooling agent on the tube side. The tube side is directly connected to the waste heat rejection loop, dissipating the system's thermal energy into space through radiation. The recuperator model (Fig. 10) resembles the gas cooler, with both sides utilizing He-Xe gas for efficient heat exchange.

4. Verification and validation

The program's main structure, as depicted in Fig. 11. Fig. 12 illustrates the computational process of the system program. The numerical

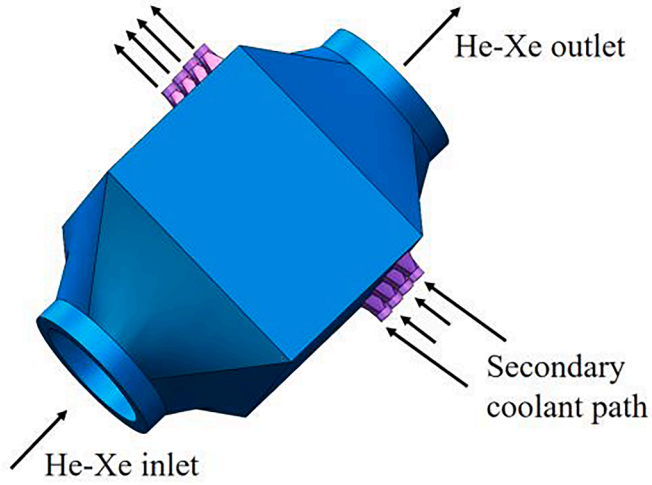


Fig. 9. Gas cooler physical model.

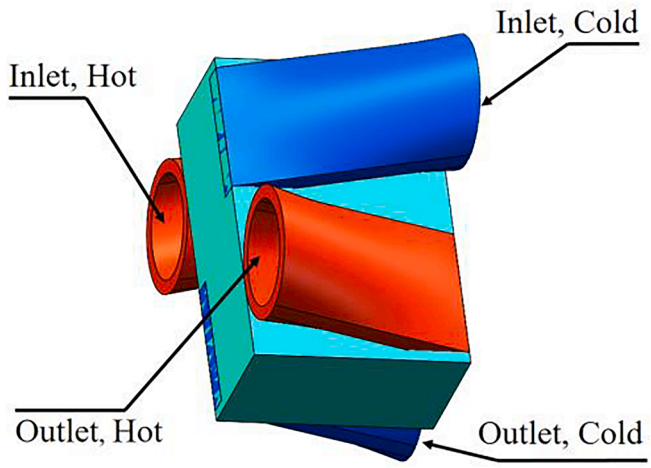


Fig. 10. Recuperator physical model.

methods of the steady state and transient are the same, the difference lies in the setting of boundary conditions. Under constant boundary conditions, the values of the system parameters after stabilization during the calculation process are used as the steady-state calculation results.

Due to the lack of overall experimental research data on core heat transfer, this work compares the program calculation results with the calculation results of CFD software with more sophisticated modeling and higher calculation accuracy. The simulation conditions and calculation data are from Ref. [17].

As shown in the Fig. 12, the distribution of the fuel core block and coolant temperature along the axial direction calculated by the program is not much different from that of the CFD program, and the relative error is within 10 %, which shows the correctness of the heat conduction and convection heat transfer models implanted in the program.

The axial distribution of the fuel core and coolant temperature calculated by the program, as depicted in Fig. 13, exhibits a relative error of less than 10 % compared to that obtained from the CFD program. The sources of error are mainly the following:

(1) CFD simulations typically employ fine grids and high-order precision methods, whereas custom programs may adopt coarser grids or simpler algorithms.
(2) Dissimilarities exist between the core power distributions in the program and those derived from CFD simulations.
(3) The presence of reflectors on both sides of the core influences neutron diffusion and heat transfer processes, reflectors tend to decelerate temperature gradients.

The heat conduction and cooling process of the core are closely interrelated during transient processes, the heat transfer capacity of the He-Xe gas directly affects the temperature field and dynamic changes during the transient process. Although the entire transient simulation process cannot be directly verified by experimental data, the accuracy of the heat transfer model can be ensured through Nu number verification to ensure that the heat transfer process of the cooling gas in the simulation is physically real.

The following is a comparative chart illustrating the heat transfer relationship utilized in the program and the experimental data results.

The correlation is typically derived under specific geometric conditions, however, actual experiments may involve different boundary conditions, including wall roughness and inlet flow distribution, which can introduce additional errors. Although there are certain inaccuracies in the Kays and Genielinski correlations within various Re number ranges, overall trends align reasonably well with experimental data while maintaining an acceptable level of error. The Fig. 14 demonstrates that these correlations effectively capture the heat transfer law and exhibit a relatively stable range of error, providing a certain degree of reliability when predicting heat transfer performance during cold reactor transient simulations.

5. Result and discussion

5.1. Positive reactivity insertion accident analysis

Reactor power is primarily influenced by reactivity changes. Unexpected reactivity insertion resulting from control system operation failure can lead to a rapid escalation in reactor power, jeopardizing the safety of fuel elements. In this section, positive reactivity insertion accidents (PRIA) of different severities are simulated to analyze the transient response characteristics of a gas-cooled reactor integrated with a He-Xe Brayton cycle system under PRIA. Given that the effects of PRIA on all four Brayton loops are similar, the results for a single Brayton cycle are presented.

The core temperature remains within a safe range as long as the introduced reactivity does not exceed 0.152$, as shown in Fig. 15. Below, the calculation and analysis results of introducing 0.13$ positive reactivity are given to explore the impact of PRIA on the efficiency of the coupling system without protection measures.

Fig. 16 illustrates the variation in core reactivity during the PRIA process, while Fig. 17 shows the transient response of reactor power and flow rate. The insertion of positive reactivity leads to a rapid surge in reactor power to 4.74 MWt (139 % of nominal power) within 15 s, accompanied by a significant increase in core temperature. Subsequently, various materials exhibit negative temperature reactivity feedback effects that restrict further escalation in power, with fuel reactivity feedback being predominant followed by cladding.

After 144 s, the total core reactivity returns to 0, and the reactor re-enters a critical state with a power level of 3.75 MWt (110 % of the nominal power). Upon reaching steady-state operation again, the turbine inlet temperature (reactor outlet temperature) is elevated compared to pre-accident transient conditions due to increased heat extraction by the turbine from the gas. Consequently, the reactor power does not revert back to its initial level.

The power the turbine extracts from the fluid is defined by the following equation:

$$P = \frac{\eta m \Delta P}{\rho} \tag{27}$$

Where η is the turbine efficiency, m is the coolant fluid mass flow rate, ΔP

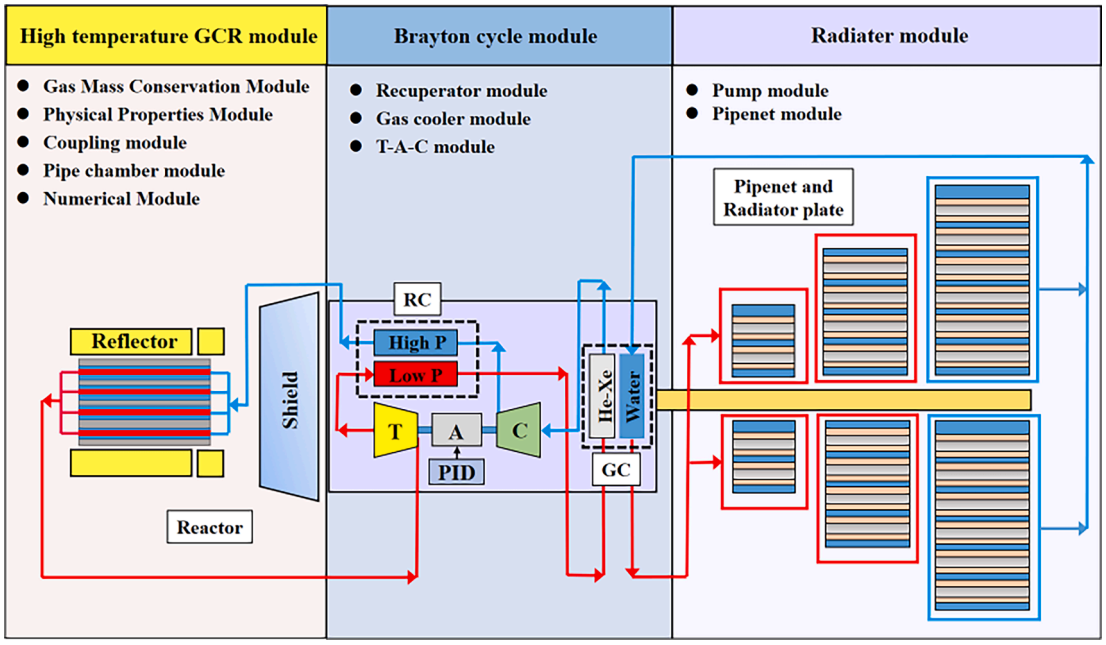


Fig. 11. Building blocks of the program.

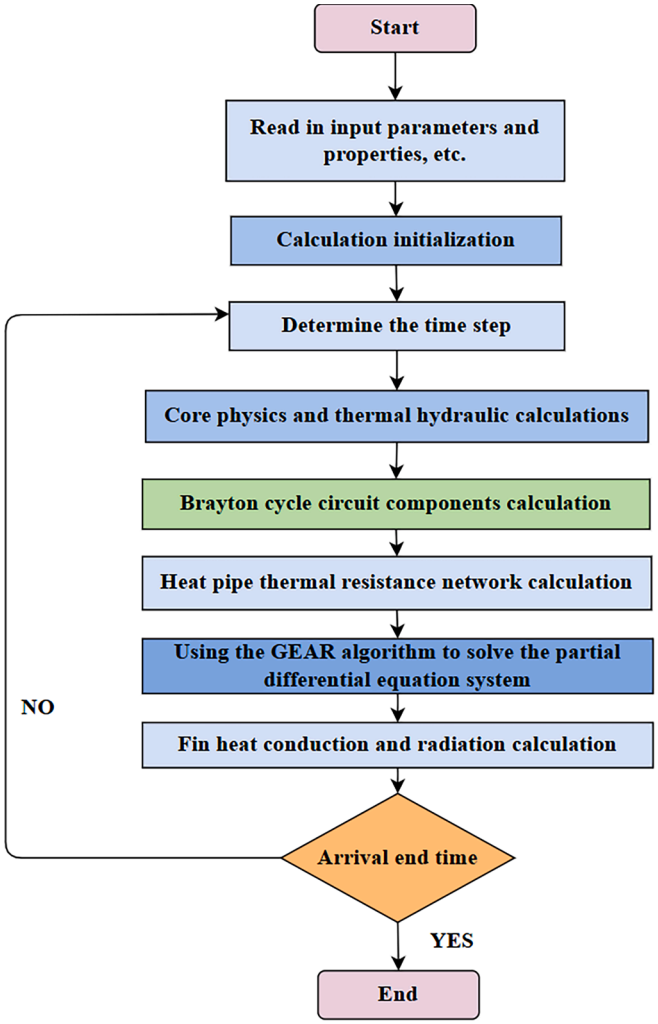


Fig. 12. flow chart of the program.

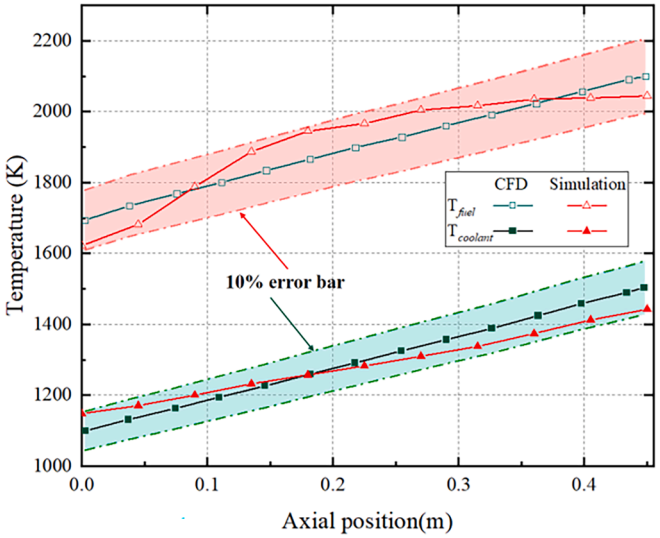


Fig. 13. Results comparison between the program code and CFD simulation.

is the pressure drop across the turbine, ρ is the average density in the turbine.

The temperature hot spot of the core and fuel cladding are depicted in Fig. 18, while Fig. 19 illustrates the changes in the average reactor temperature. It can be observed that the fuel temperature reaches its peak at 38 s, whereas the cladding temperature peaks at 64 s. Both temperatures exhibit a slight time lag compared to the peak time of core power due to thermal inertia resulting from energy deposition within the fuel.

During the entire PRIA process, the fuel temperature undergoes a gradual increase from 2125 K, reaching its peak at 2347 K, and eventually stabilizing at 2287 K. Similarly, the maximum temperature of the fuel cladding rises from 1761 K, peaks at 1919 K, and then stabilizes at 1895 K. It is noteworthy that the cladding exhibits greater vulnerability compared to the fuel. The peak temperature of the fuel maintains a safety margin of 126 K below its limit, while for the cladding this margin is only 31 K.

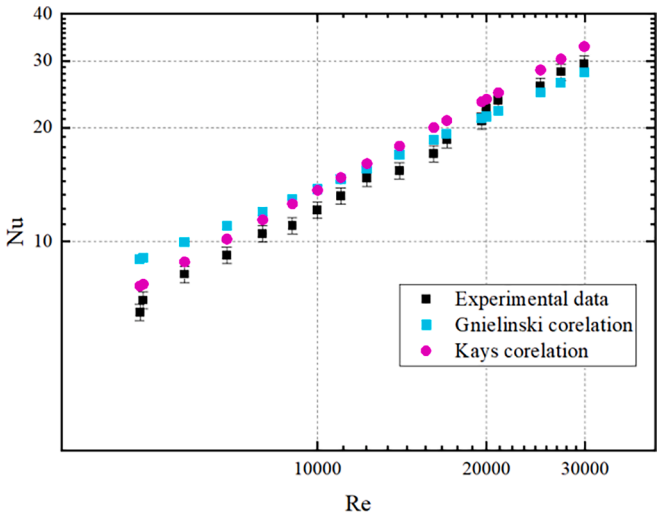


Fig. 14. Comparison of Nu experimental data and heat exchange correlations used in the program.

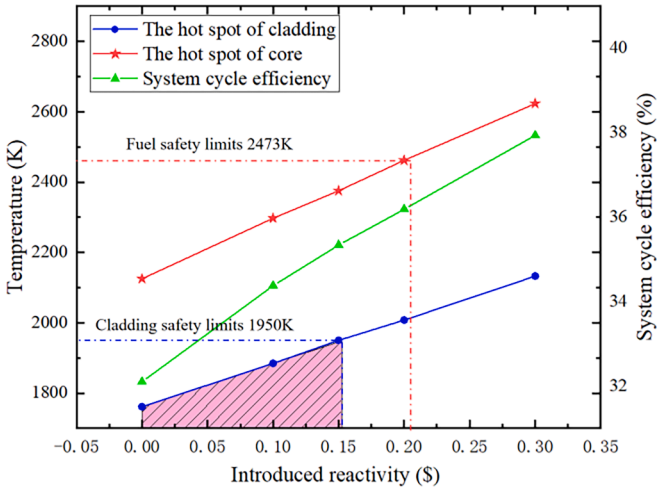


Fig. 15. Safety margins of accident conditions with different reactivity.

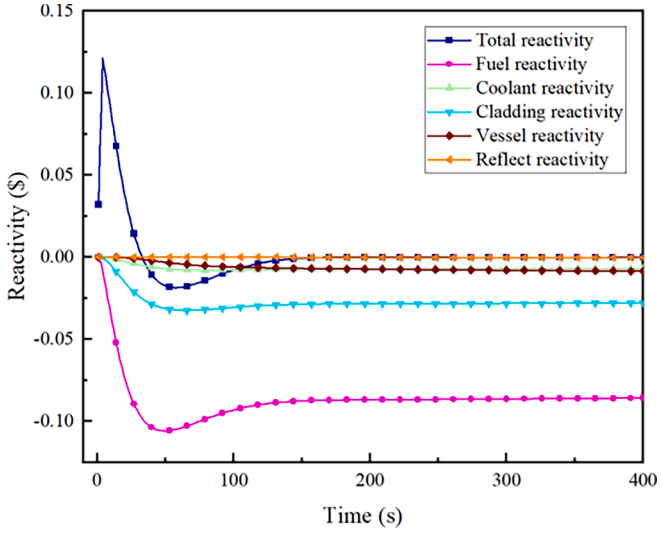


Fig. 16. The core reactivity changes after introducing 0.13$.

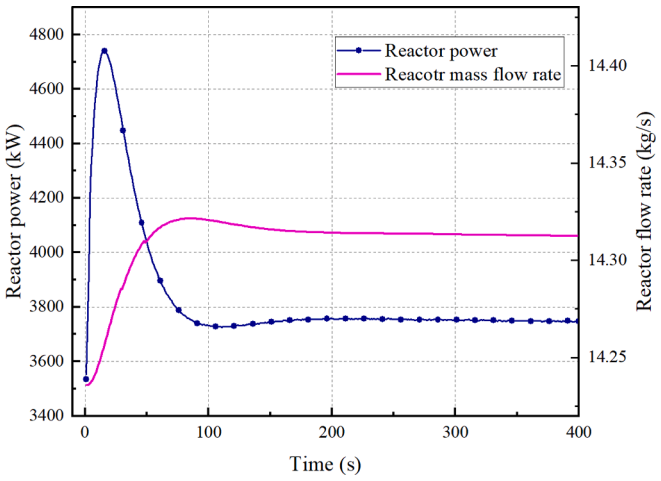


Fig. 17. The reactor power and flow changes after introducing 0.13$.

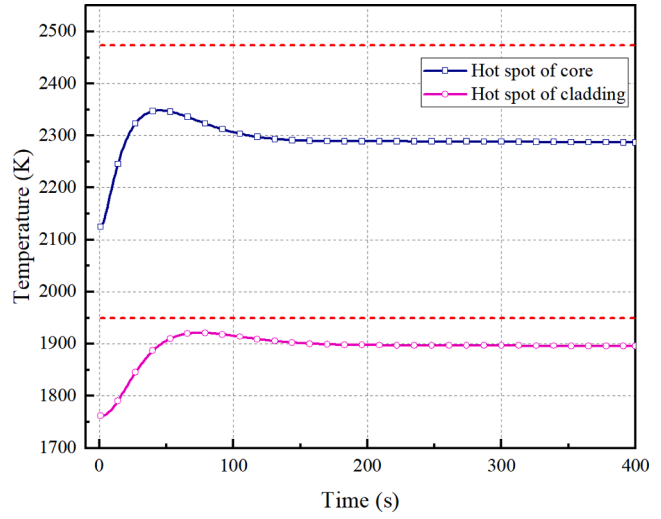


Fig. 18. The temperature hot spot of the core and cladding after introducing 0.13$.

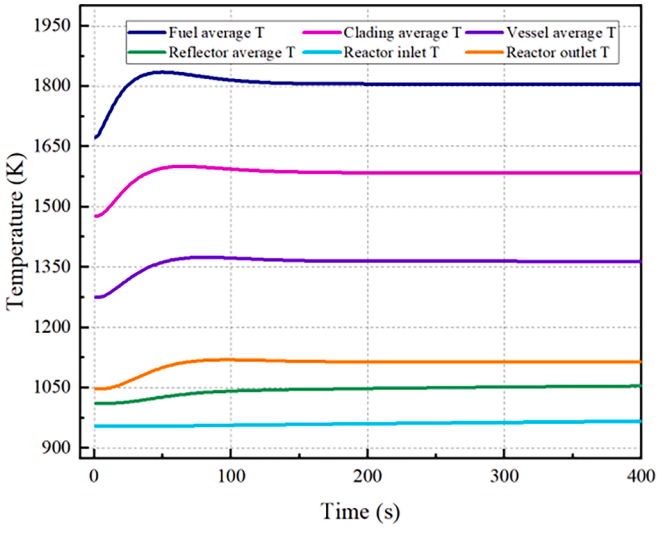


Fig. 19. The average temperature of the core after introducing 0.13$.

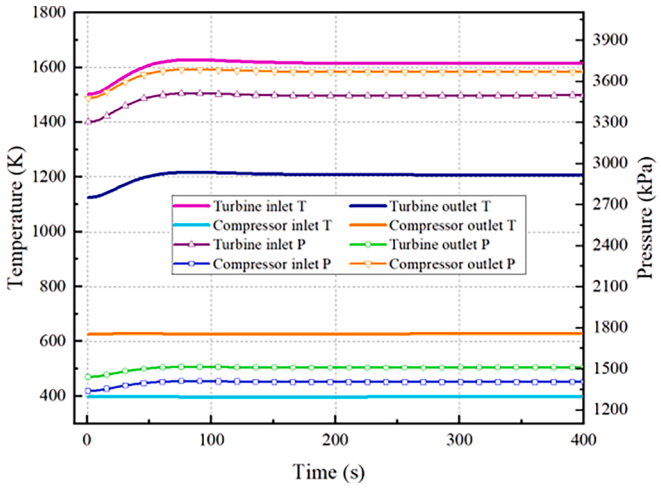


Fig. 20. Turbine and Compressor temperatures and pressures after introducing 0.13$.

The temperature and pressure response of the turbine and compressor is illustrated in Fig. 20. As the overall core temperature increases due to a constant coolant flow, the turbine inlet temperature also increases accordingly. For He-Xe gas, this rise in temperature not only leads to an increase in pressure at the turbine but also a decrease in He-Xe gas density at both the turbine inlet and outlet. According to Eq.(27), when the average density of He-Xe gas at the turbine decreases by 1.1 %, there is a corresponding increase in turbine power. This change can be observed in Fig. 21, where the turbine power rises from 695 kWt to 755 kWt. Meanwhile, the torque required for compressor operation remains essentially unchanged, with its power increasing only from 422 kWt to 426 kWt. To maintain a constant TAC shaft speed, PMAD must increase generator power from 273 kWe to 327 kWe (121 % of rated value), resulting in an overall increase in power production.

In the new steady state conditions, reactor power reaches up to 3.75 MWt while total electrical output hits at around 1.31MWe, the overall efficiency rises from its original level of 32.1 % to 34.8 %.

The fuel cladding temperature will exceed the safety limit of 1950 K when the positive reactivity insertion exceeds 0.152$. In this work, control rod insertion into the core is adopted as a safeguarding measure to ensure the integrity of fuel elements remains uncompromised. The

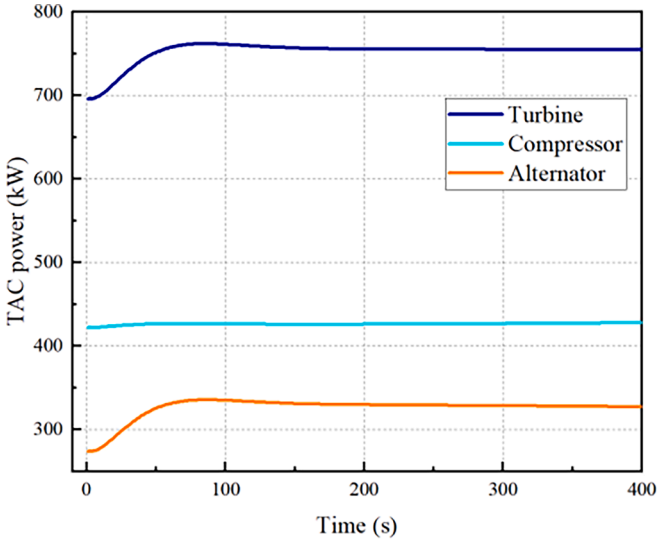


Fig. 21. The power of turbine-alternator-compressor (TAC) after introducing 0.13$.

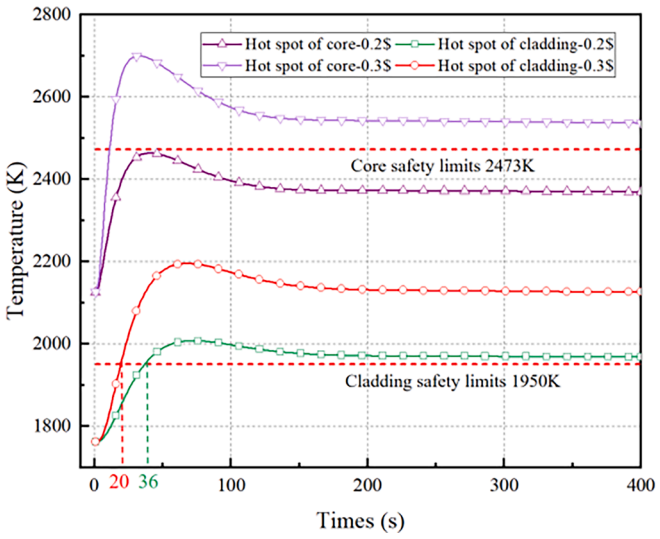


Fig. 22. The temperature hot spot of the core and cladding after introducing 0.2$ and 0.3$.

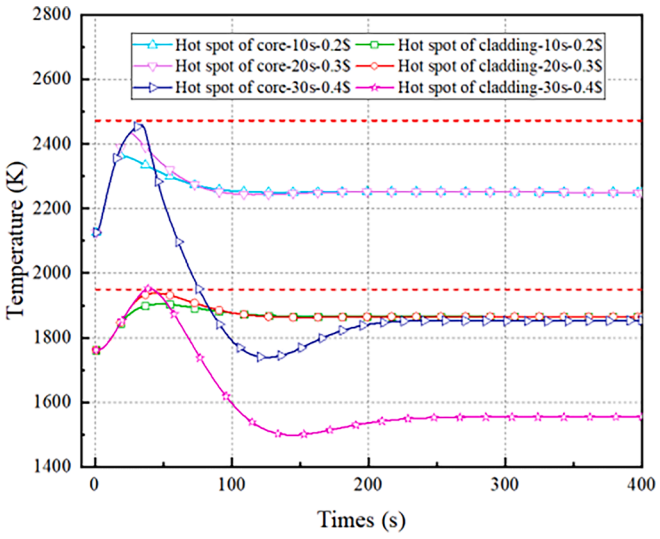


Fig. 23. The temperature hot spot of the core and cladding after introducing 0.2$ (take protective measures at 10 s, 20 s, and 30 s).

temperature hot spot of the core and cladding after positive reactivity insertion is illustrated in Fig. 22. When introducing 0.2$ positive reactivity, negative reactivity could be introduced at intervals of 10 s, 20 s, and 30 s respectively; whereas when introducing 0.3$ positive reactivity, negative reactivity could be introduced at 10 s.

Fig. 23 shows the temperature hot spot of the core and fuel cladding when protective measures are taken at 10 s, 20 s, and 30 s after the introduction of 0.2$ positive reactivity. The later the protective measures are taken, the larger negative reactivity needs to be introduced to ensure the safety of the cladding. The transient response simulation results of the reactor and Brayton cycle are presented next to investigate the impact of reactivity insertion on the system's cycle efficiency. Specifically, a negative reactivity of 0.4$ is introduced at 30 s after an initial positive reactivity accident scenario of 0.2$.

The core reactivity variation is illustrated in Fig. 24 following the introduction of negative reactivity, while Fig. 25 depicts the transient response of reactor power and flow rate. Prior to reaching the safety limit for fuel cladding (at 36 s), the core power experiences a sharp increase, peaking at 5.67 MWt (166 % of the rated value) at 14 s.

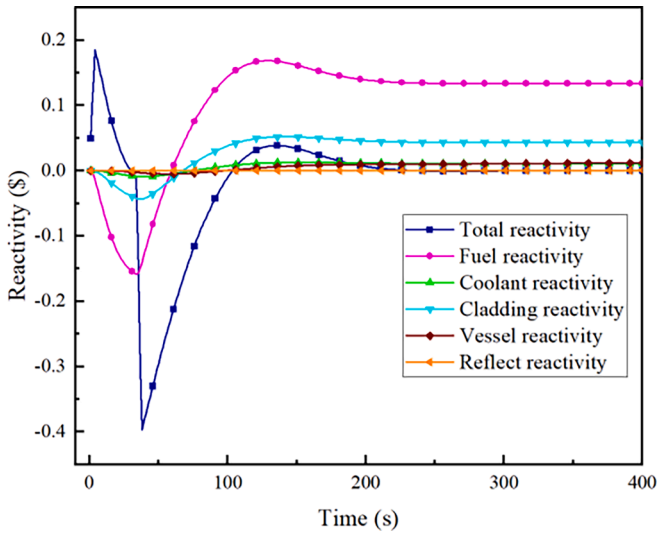


Fig. 24. The core reactivity changes after taking protective measures.

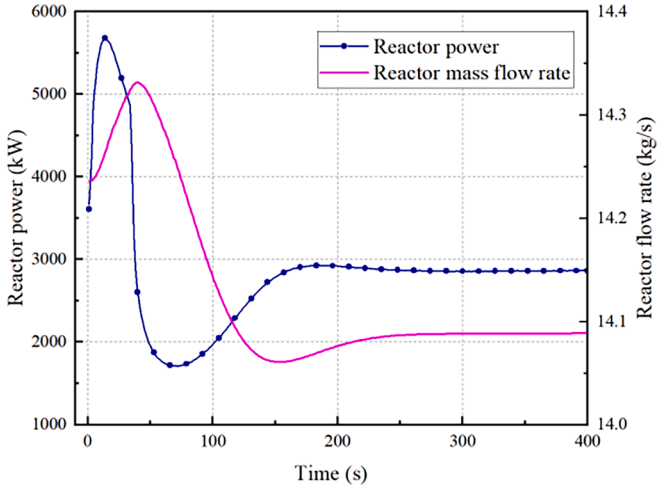


Fig. 25. The reactor power and flow changes after taking protective measures.

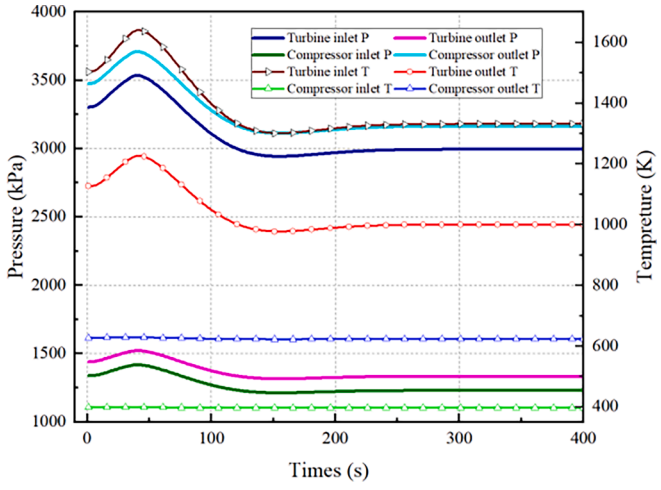


Fig. 26. Turbine and compressor temperatures and pressures after taking protective measures.

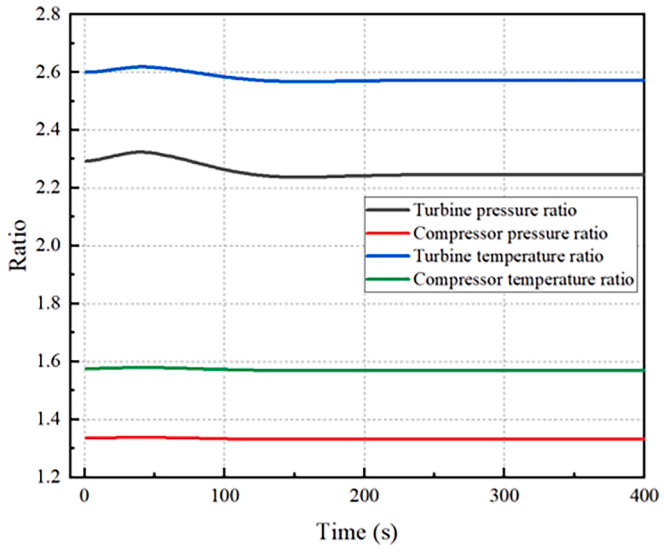


Fig. 27. Pressure and temperature ratio of turbine and compressor after taking protective measures.

Subsequently, the negative temperature reactivity feedback effects from each material impede further power escalation. Upon introducing a negative reactivity of 0.4$, there is a significant decrease in total core reactivity, resulting in a further reduction in core power.

The changes in turbine and compressor pressures and temperatures after implementing protective measures are depicted in Fig. 26, while Fig. 27 illustrates the variations in turbine and compressor pressure ratio and temperature ratio. Upon returning to steady-state operation, both the inlet and outlet pressures of the turbine and compressor will be lower compared to pre-accident transient values. Throughout the process of introducing reactivity into the system, there is no significant variation observed in the flow rate of the reactor.

Fig. 28 shows the variation of TAC power after implementing protective measures. The turbine power reaches its peak at 768.32 kWt at 41 s, starting from an initial value of 695.67 kWt, and stabilizes at 605.3 kWt. Meanwhile, the generator power stabilizes at 192.38 kWt from an initial value of 273.55 kWt. Introducing a positive reactivity of 0.2$ increases the overall cycle efficiency from an initial value of 32.1 % to 36.18 %. However, implementing protective measures (introducing a

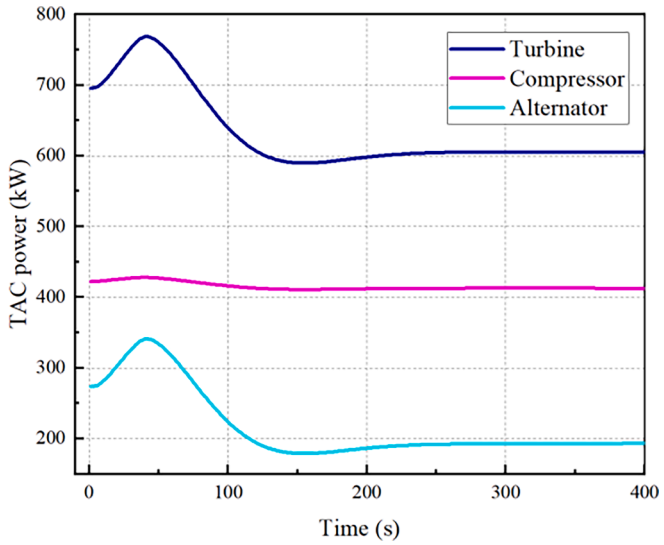


Fig. 28. The power of turbine-alternator-compressor (TAC) after taking protective measures.

negative reactivity of 0.4$) results in reactor power stabilizing at 2.87 MWt and total electrical power reaching only 769 kWe, with efficiency decreasing to 26.8 %.

The flow rate and temperature of He-Xe gas are key factors affecting the system's electricity generation while ensuring safety performance is maintained. When positive reactivity is introduced, turbine power increases as the density of He-Xe gas decreases accordingly; meanwhile, compressor torque remains relatively constant due to small variations in the flow rate of He-Xe gas. Consequently, this leads to an increase in electricity generation for the Brayton system. On the other hand, when negative reactivity is introduced, the flow rate of He-Xe gas still remains stable while its density increases, resulting in a decrease in turbine power and electricity generation for the system.

5.2. Turbine-Alternator-Compressor(TAC) shaft speed drop

In the Brayton cycle of a space reactor, the turbine shaft speed is typically adjusted to accommodate varying power requirements during the mission. In such scenarios, it becomes necessary to regulate the turbine shaft speed in order to maintain system performance stability. The steady-state results are utilized as initial conditions in calculations, with the shaft speed set at its rated value of 1170 rad/s. At 0 s, the shaft speed is reduced to 80 % and 70 % of its rated value respectively, aiming to explore the safety operating boundaries of the reactor.

The transient response characteristics of reactor power and flow rate are illustrated in Fig. 29. As turbine speed decreases, there is a corresponding reduction in coolant mass flow rate through it which leads to insufficient cooling of the core resulting in rapid increase in fuel assembly temperature followed by an elevation in fuel cladding and coolant temperatures. Fig. 30 illustrates how hot spot vary for both core and cladding during this transient response period. When the shaft speed decreases to only 70 % of its rated value, the cladding temperature reaches a hot spot exceeding 1992 K, which surpasses the safety limit. Therefore, it becomes imperative to take protective actions within 37 s (the time at which the fuel cladding reaches its safety limit temperature).

Fig. 31 shows the transient response of the hot spot of the core and fuel cladding after take protective measures. Within the first 10 s after the decrease in shaft speed, only 0.1$ negative reactivity is needed to reduce the maximum fuel cladding temperature to within the safety limit. Based on Fig. 32 and Fig. 33, it can be observed that the introduction of at least 0.2$ and 0.3$ negative reactivity is required when protective measures are implemented at 20 s and 30 s, respectively. When 0.3$ negative reactivityis introduced at 30 s, although the peak

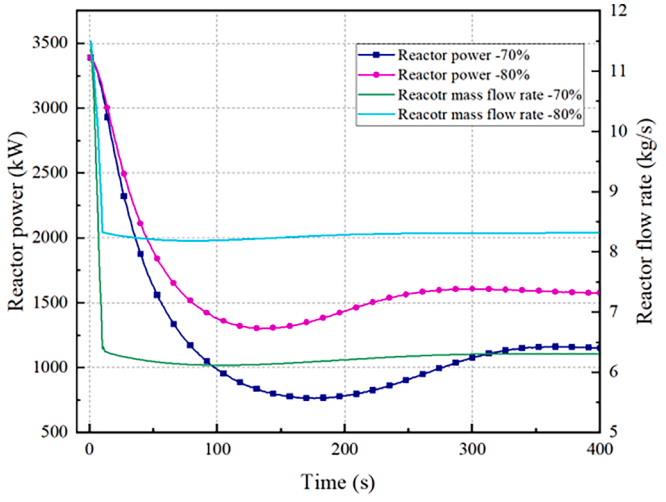


Fig. 29. Transient response of reactor power and flow rate after changing shaft speed.

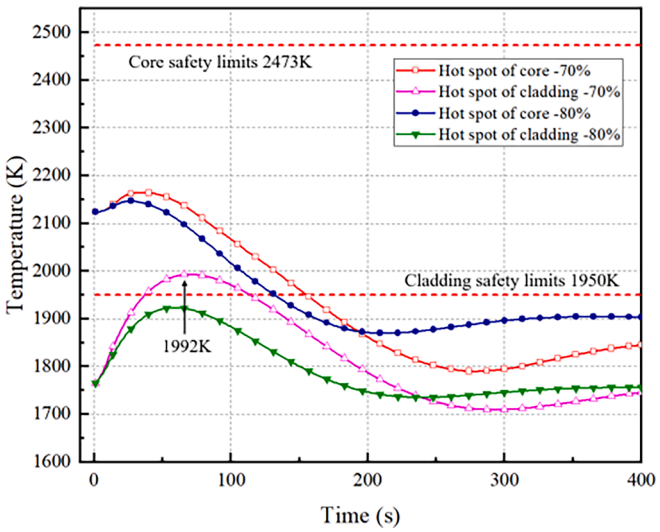


Fig. 30. Transient response of the hot spot of core and cladding after changing shaft speed.

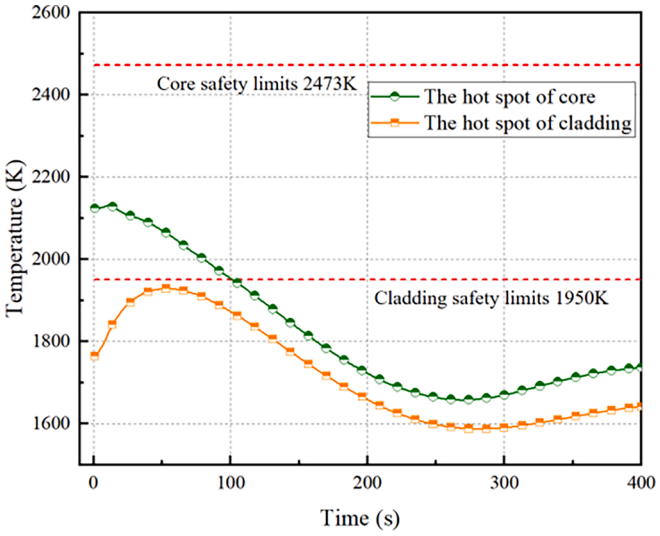


Fig. 31. The hot spot of the core and fuel cladding after the introduction of 0.1 $ negative reactivity at the 10 s.

temperature approaches the cladding safety limit of 1950 K, its duration is short enough to maintain cladding integrity.

The demand for shaft speed in practical applications is characterized by complexity and variability, often necessitating continuous adjustments to meet specific requirements. The following calculations and analysis present the corresponding results for the shaft speed at "0s-50s-260s," which correspond to the rated speeds of "100–90–80 %.

The transient response of reactor power and flow rate is depicted in Fig. 34 as the shaft speed continuously decreases to 80 % of the rated speed. Fig. 35 and Fig. 36 illustrate the variations in core temperature and reactivity feedback of different materials under continuous variable shaft speed conditions. Initially, there is a rapid increase in core temperature due to the reduction in He-Xe gas mass flow and decreased heat exchange in the intermediate heat exchanger. This leads to corresponding increases in fuel cladding and coolant temperatures. However, subsequent negative temperature reactivity feedback effects from various materials, suppress further temperature rise and eventually cause a decrease in reactor core temperature.

The pressure and temperature variations within the turbine and compressor are illustrated in Fig. 37 and Fig. 38. Decreasing the turbine

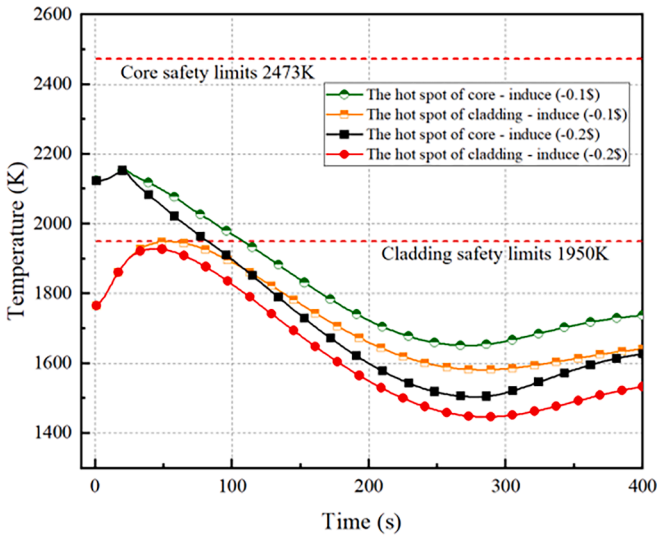


Fig. 32. The hot spot of the core and fuel cladding after the introduction of 0.1 $ and 0.2$ negative reactivity at 20 s.

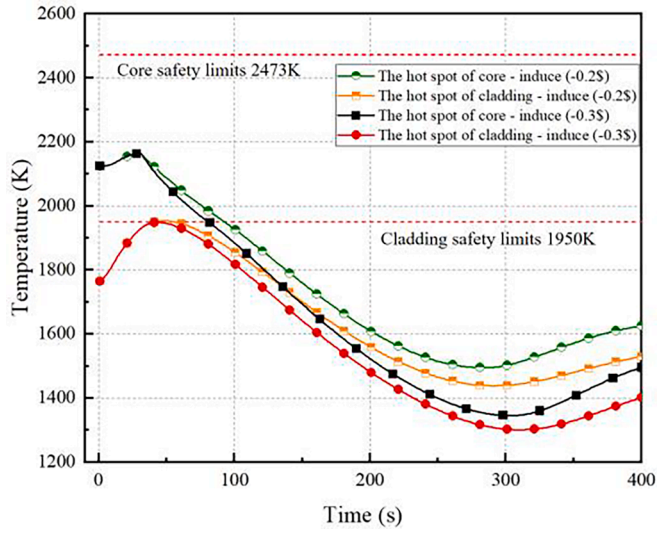


Fig. 33. The hot spot of the core and fuel cladding after the introduction of 0.2 $ and 0.3$ negative reactivity at 30 s.

and compressor temperature ratios and pressure ratios further leads to an increase in turbine inlet and outlet temperatures, as well as a decrease in pressure. Simultaneously, the compressor outlet temperature decreases while the pressure increases.

The power variation of the turbine-alternator-compressor over time is illustrated in Fig. 39. The decrease in shaft speed leads to a reduction in the flow rate of He-Xe gas passing through the turbine, there is a significant decrease in power output, while compressor power is directly proportional to loop flow rate.

The relationship between the power of a compressor and the flow rate of the refrigerant can be explained by the principle of energy conservation. The primary function of a compressor is to increase the pressure of the refrigerant (such as helium) from a low-pressure region to a high-pressure region, which necessitates power consumption. As the flow rate of the refrigerant increases, there is an accompanying increase in gas volume that needs to be handled by the compressor, resulting in higher power requirements for maintaining consistent compression ratio and efficiency. Furthermore, an elevated flow rate may lead to variations in temperature and pressure drop during compression, thereby

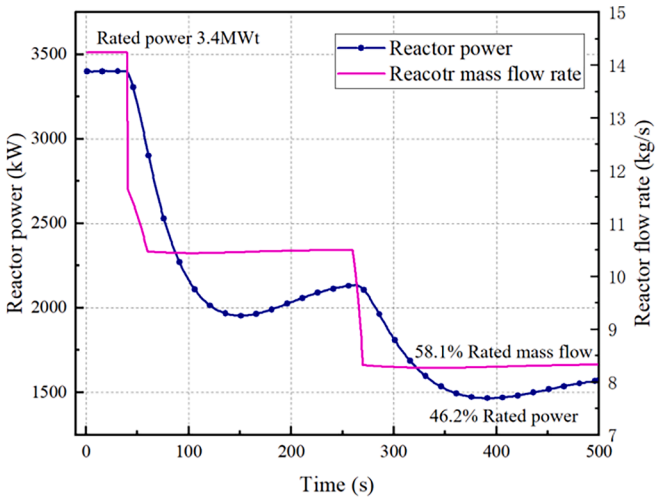


Fig. 34. Transient response of reactor power and flow rate after continuous shaft speed change.

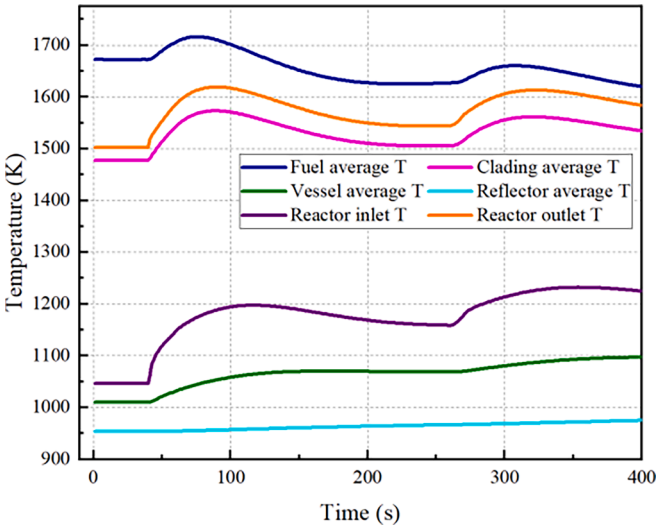


Fig. 35. Reactor average temperature after continuous shaft speed change.

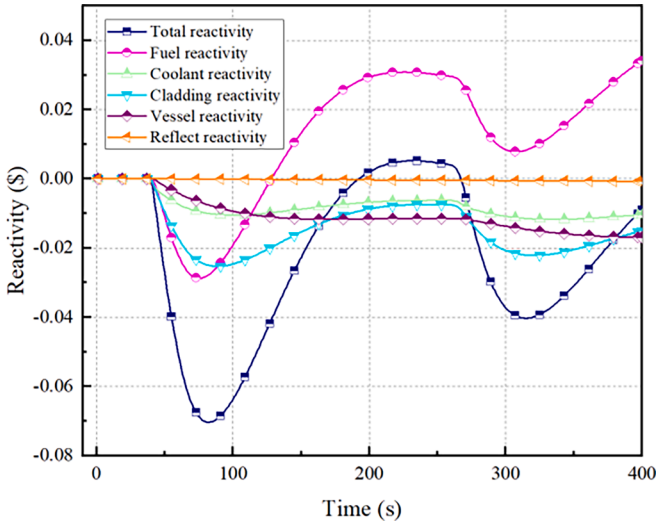


Fig. 36. Reactivity feedback of core materials after continuous shaft speed change.

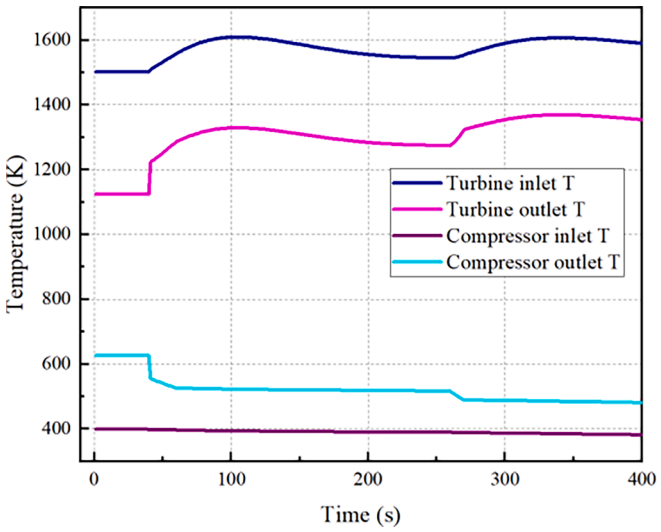


Fig. 37. Transient response of turbine and compressor temperature after continuous shaft speed change.

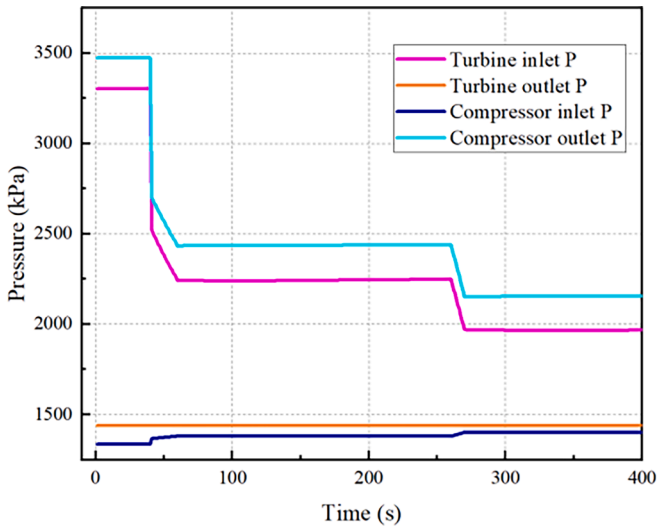


Fig. 38. Transient response of turbine and compressor temperature after continuous shaft speed change.

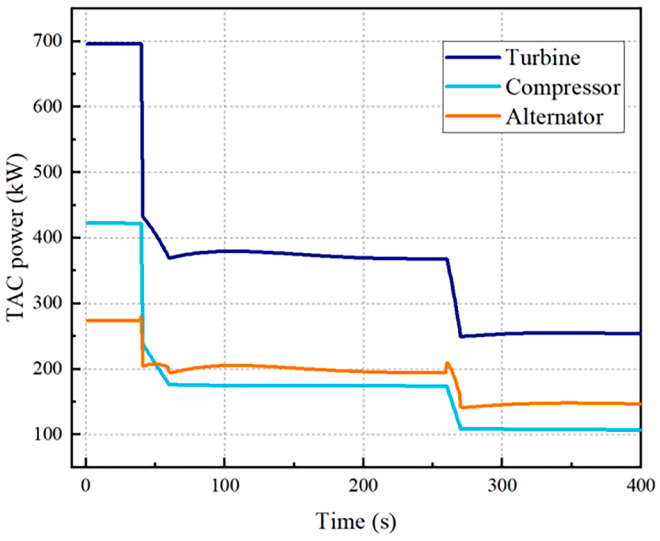


Fig. 39. The power TAC after continuous shaft speed change.

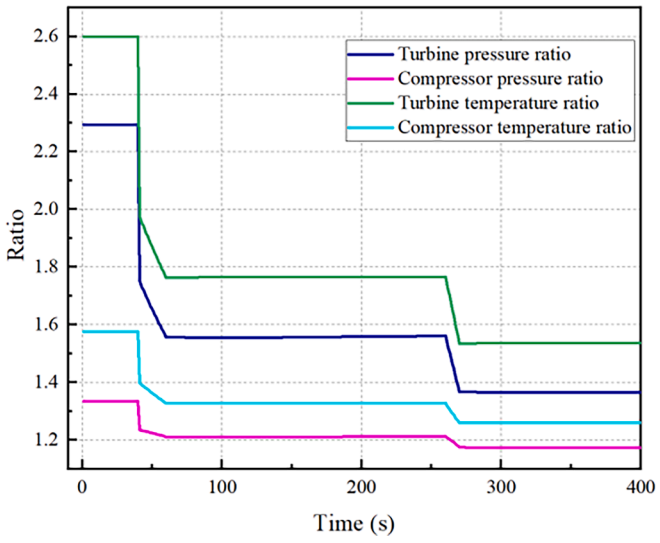


Fig. 40. Pressures ratio and temperatures ratio of turbine and compressor after continuous shaft speed change.

further augmenting power demands.

In addition, as shown in Fig. 40, the decrease in shaft speed leads to a reduction in the temperature ratio and pressure ratio of the turbine and compressor, which can be explained by the principles of gas dynamics. As the shaft speed decreases, there is a corresponding decrease in fluid flow rate through both the turbine and compressor, resulting in a decline in energy conversion efficiency. In turbines, lower speeds lead to less expansion of gas within the turbine, causing a decrease in temperature ratio and pressure ratio. Similarly, at lower speeds, compressors experience reduced effectiveness during the compression process, leading to decreased output pressure and temperature.

5.3. Loop coolant flow rate reduction

The Loss of Flow Accident (LOFA) encompasses various scenarios, including partial or complete loss of flow, compressor seizure, and compressor shaft breakage. This work aims to simulate a sudden reduction in flow within the Brayton module of the gas-cooled space reactor. Under such accident conditions, there is a rapid increase in core temperature, which may lead to the fuel cladding exceeding safety limits. The following calculation determines the hot spot of the fuel cladding when the system experiences a sudden drop in flow, thereby exploring safety boundaries for different degrees of accident conditions.

Fig. 41 depicts the hot spot of fuel cladding temperature of the core and the thermal cycle efficiency of the system (expressed as a percentage of rated power) during a flow reduction accident. At 52 % of rated flow, the hot spot of fuel cladding temperature reaches its safety limit, causing a decrease in system cycle efficiency to 11 %. During this process, if system flow falls below 50 %, there is an imminent risk of fuel cladding melting. Protective measures must be taken within 43 s from when peak temperature is reached.

The calculation results for reducing the system flow to 60 % of its rated flow are presented herein. Fig. 42 and Fig. 43 depict the temporal variations in negative reactivity for different core materials and the core temperature throughout the process. Additionally, Fig. 44 illustrates the transient response of reactor power and core flow rate.

From the Fig. 44, it is evident that the rapid increase in fuel assembly temperature is a result of the sharp decrease in coolant flow rate without timely adjustment of reactor power. However, due to the negative reactivity feedback effect of the fuel, introduced negative reactivity

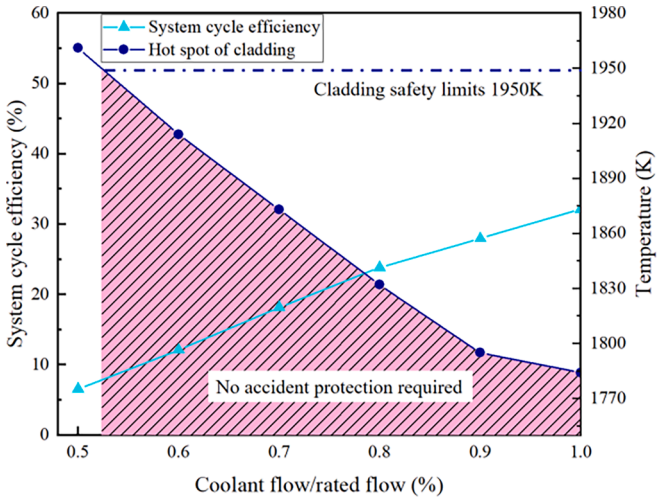


Fig. 41. Safety margins of accident conditions when the flow rate is reduced to different degrees.

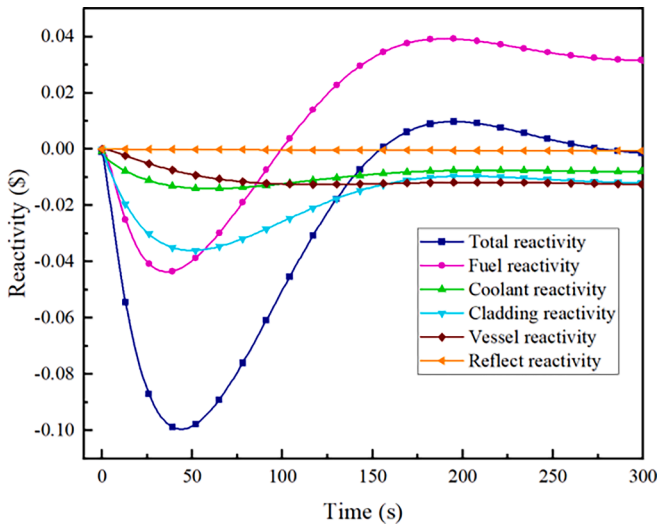


Fig. 42. Reactor reactivity feedback changes after flow drops.

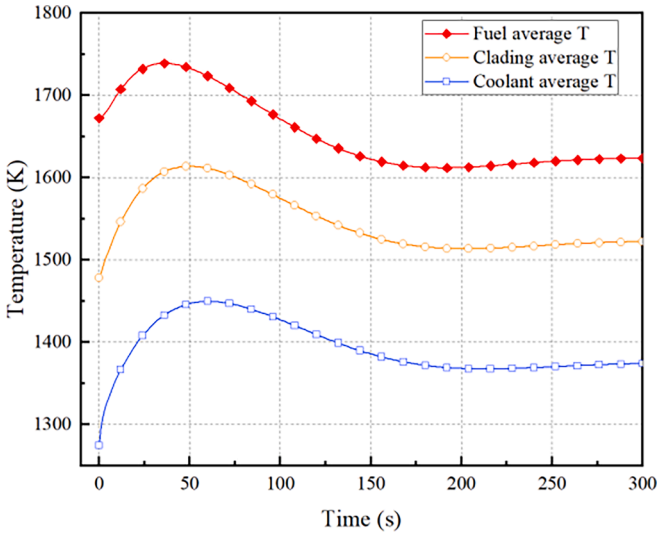


Fig. 43. Core average temperature changes after flow drops.

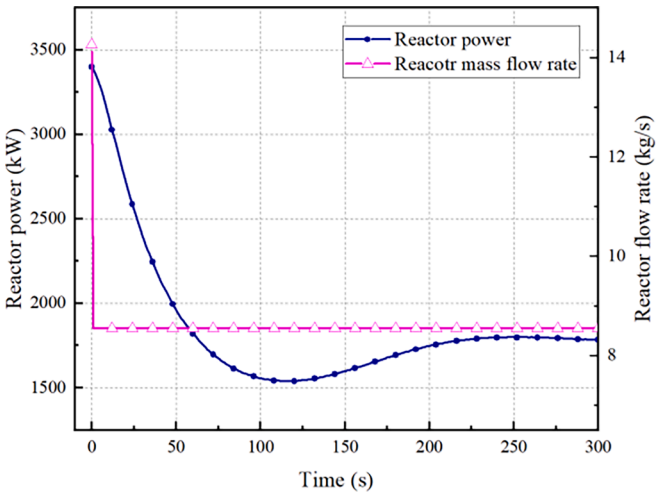


Fig. 44. Reactor flow and power changes after flow drops.

peaks at 1738 K at 38 s and gradually diminishes under the overall negative reactivity effect of core materials. Simultaneously, reactor power continues to decline following the loss of flow accident, further reducing core temperature. Subsequently, with the introduction of reactivity, reactor power continues to decrease while fuel temperature starts to decline as well. Consequently, the absolute value of negative reactivity introduced by fuel feedback gradually decreases and eventually reaches 0 around 250 s when the reactor attains a new stable state.

The temperature and pressure changes of the turbine and compressor over time are illustrated in Fig. 45 and Fig. 46. A significant drop in the loop flow rate, coupled with a constant set shaft speed, leads to an elevation in both turbine inlet and outlet temperatures as well as an accompanying increase in system-wide pressure.

The changes in the temperature ratio and pressure ratio of the turbine and compressor are illustrated in Fig. 47. As the coolant flow rate decreases, it impacts the input and output pressures of both the turbine and compressor. The reduced flow rate causes the compressor to process less gas at the same speed, resulting in an increased pressure ratio. This is because the compressor needs to maintain a consistent pressure output under lower flow conditions. Similarly, due to decreased input heat, the turbine's output power also diminishes, leading to a corresponding increase in pressure ratio.

Fig. 48 depicts variations in TAC shaft power when there is a sudden

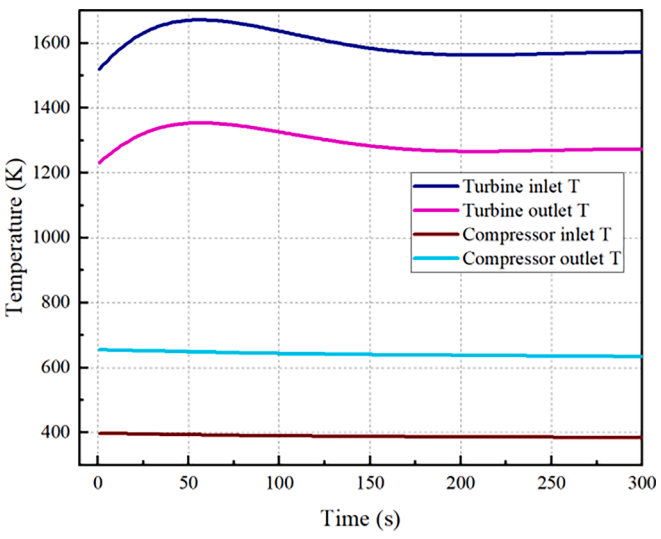


Fig. 45. Turbine and compressor temperature changes after flow drops.

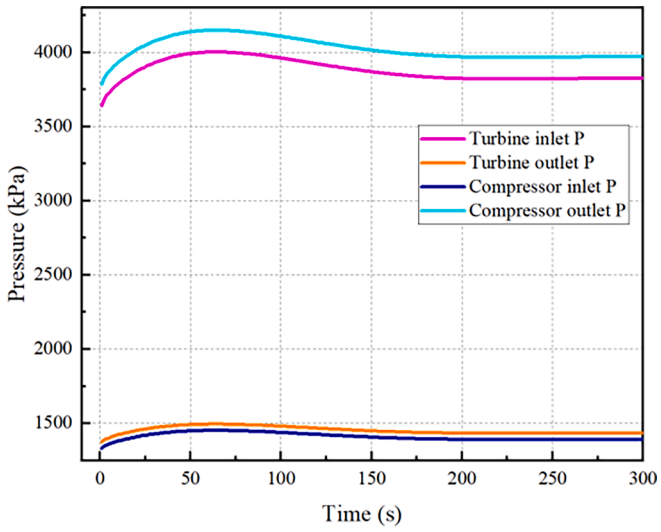


Fig. 46. Turbine and compressor pressure changes after flow drops.

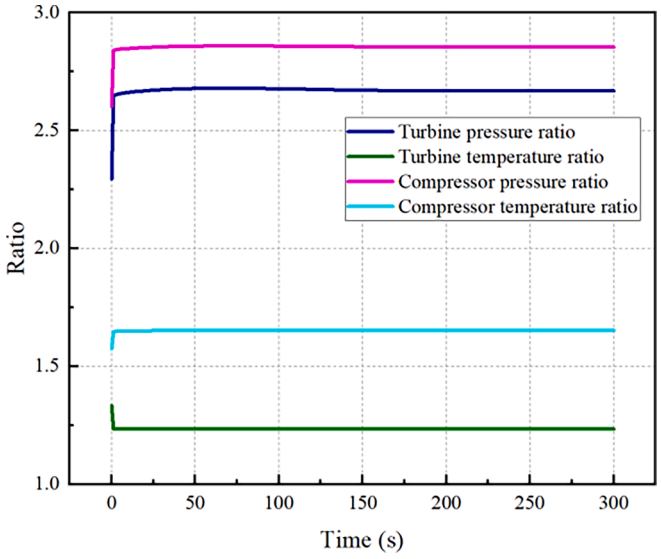


Fig. 47. Turbine and compressor temperature ratio and pressure ratio changes after flow drops.

decrease in system flow rate. A reduction in coolant flow rate results in diminished cooling effect within the reactor, leading to an increase in reactor temperature and potential limitations on reactor power output. Simultaneously, due to reduced heat provided by the reactor, TAC shaft power output also decreases. Reactor power is reduced to 1.78 MWt while total electrical power drops from 1096 kWe to 216 kWe. In the Brayton cycle, a higher pressure ratio typically leads to enhanced thermal efficiency as it enables better utilization of input heat energy for work conversion. However, when the coolant flow rate is reduced and the pressure ratio becomes excessively high, although there may be theoretical improvement in cycle efficiency, in reality, due to decreased efficiency of both the turbine and compressor, overall thermal cycle efficiency will decrease, a decline in system cycle efficiency from 32.1 % to 12.13 %.

6. Conclusion

The present study employs a developed thermal–hydraulic system analysis program to perform safety characteristic analyses of typical accident scenarios for megawatt-level space gas-cooled reactor power,

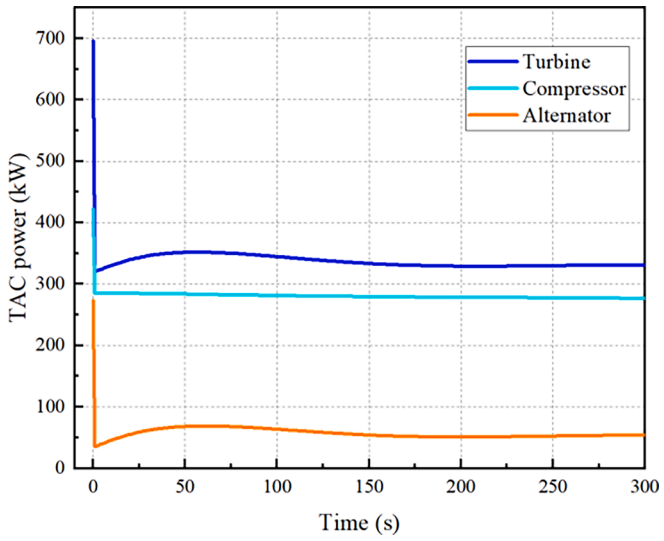


Fig. 48. TAC shaft power changes after flow drops.

encompassing positive reactivity insertion, Turbine-Alternator-Compressor (TAC) shaft speed reduction, and sudden decrease in loop coolant flow rate.

The introduction of positive reactivity into the reactor can enhance the thermal cycle efficiency of the system. Positive reactivity optimizes neutron economy within the reactor, thereby increasing the effectiveness of fission reactions and ultimately improving energy release efficiency. This power augmentation leads to greater heat generation in the reactor. Simultaneously, when the He-Xe gas carries away more heat entering the turbine, it further improves both operating temperature and heat transfer efficiency of the thermal cycle system.

In the Brayton cycle, a decrease in turbine-alternator-compressor (TAC) shaft speed generally results in a decrease in system efficiency, whereas an increase in TAC shaft speed generally results in an increase in system efficiency. This is attributed to the fact that higher speeds can optimize power output of the thermal cycle, thereby improving heat energy conversion effectiveness. However, it should be noted that continuous efficiency improvement is not guaranteed with increasing speeds; excessively high speeds may give rise to issues such as stall phenomena and thermodynamic losses within the system. Additionally, a sudden drop in coolant flow rate may cause a temperature rise in the reactor which resulting in fuel cladding melting. Within an appropriate range of speeds, there exists a positive correlation between TAC shaft speed and cycle efficiency.

When the coolant flow rate suddenly decreases in a gas-cooled reactor coupled with a Brayton cycle system, the reactor power and TAC power decrease due to the combined effects of reduced heat exchange efficiency and limited reactor power output. Simultaneously, the decrease in flow rate leads to changes in operating conditions of the system, resulting in an increase in pressure ratio for both turbine and compressor. Ultimately, this increased pressure ratio fails to effectively enhance cycle performance, leading to a decline in thermal cycle efficiency and greater energy loss. The overall temperature rise of the system also indicates a reduction in energy utilization efficiency.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

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China 2023NSCQ-BHX0243.

Data availability

Data will be made available on request.

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